KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM

(An Autonomous Institution)

Approved by AICTE, Affiliated to Anna University, Chennai.
Accredited by NBA (CSE, ECE, EEE & MECH), Accredited by NAAC with „A" Grade

KIOT Campus, Kakapalayam – 637 504. Salem Dt., Tamil Nadu, India.

M.E. / M.Tech. Regulations 2023

M.E. – Automotive Electronics

Curriculum and Syllabi

(For the Students Admitted from the Academic Year 2023 – 2024 onwards)

Version: 1 .0	**Date:** 15.12.2025

M.E. / M.Tech. / M.C.A. REGULATIONS 2023 (R 2023)
CHOICE BASED CREDIT SYSTEM AND OUTCOME BASED EDUCATION

M.E. AUTOMOTIVE ELECTRONICS

VISION OF THE INSTITUTE

- To be a world-class institution to impart value and need based professional education to the aspiring youth and carving them into disciplined world class professional who have the quest for excellence, achievement orientation and social responsibilities.

MISSION OF THE INSTITUTE

A	To promote academic growth by offering state-of-art undergraduate, postgraduate anddoctoral programs and to generate new knowledge by engaging in cutting – edge research
B	To nurture talent, Innovation, entrepreneurship, all-round personality and value system among thestudents and to foster competitiveness among students
C	To undertake collaborative projects which offer opportunities for long-term interaction with academia and industry for creating a sustainable world.
D	To pursue global standards of excellence in all our endeavors namely teaching, research, consultancy, continuing education and support functions

VISION OF THE DEPARTMENT

To produce competent Electronics and Communication Engineers by imparting quality education to meet the industry requirements and for serving the societal needs

MISSION OF THE DEPARTMENT

M1	To develop appropriate facilities for promoting research activities
M2	To inculcate leadership qualities among students for self and societal growth
M3	To nurture students on emerging technologies for serving industry needs through industry institute interface
M4	To enrich teaching learning process by transforming young minds to be resourceful engineers

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	To gain knowledge and practical skills in automotive electronics, sensors, and control systems.
PEO 2	To apply engineering principles for research, innovation, and development of modern vehicle technologies such as electric and intelligent systems.
PEO 3	To demonstrate professionalism, teamwork, and continuous learning for growth in the automotive field.

PROGRAM OUTCOMES (POs)	
PO 1	An ability to independently carry out research/investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document
PO 3	Students should be able to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program
PO 4	Understand the fundamentals involved in the Designing and Testing of Automotive electronics
PO 5	Provide Solutions Through Research to Socially Relevant Issues in Automotive Electronics
PO 6	Interact effectively with the technical experts in industry

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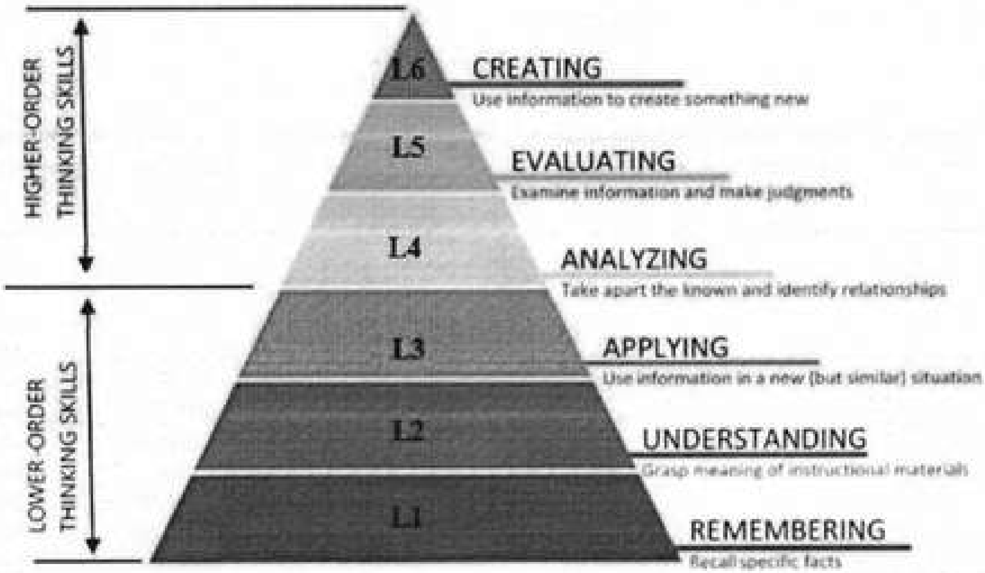
BLOOM'S TAXONOMY LEVELS AND ACTION VERBS

(A) BLOOM'S TAXONOMY LEVELS (BTL):

Bloom's Taxonomy (BT) is based on the belief that learners must first acquire basic foundational knowledge about a subject before progressing to more complex types of thinking, such as analysis and evaluation. Bloom's Taxonomy levels help faculty to guide students through the learning process, from fundamental remembering and understanding to more complex evaluating and creating.



At KIOT, Curriculum Design, Delivery and Assessment (CDDA) are carried out based on the Blooms' Taxonomy Levels (BTL). Its organized set of objectives helps teachers to plan and deliver appropriate instruction, design valid assessment tasks and schemes, and ensure that instruction and assessment are aligned with the objectives.

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(B) BLOOM'S TAXONOMY ACTION VERBS

I. Remembering	II. Understanding	III. Applying	IV. Analyzing	V. Evaluating	VI. Creating
Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answers.	Demonstrate understanding of facts and ideas by organizing, comparing, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.	Present and defend opinions by making judgments about information, validity of ideas, or quality of work based on a set of criteria.	Compile information together in a different way by combining elements in a new pattern or proposing new Solutions.
• Choose • Define • Find • How • Label • List • Match • Name • Omit • Recall • Relate • Select • Show • Spell • Tell • What • When • Where • Which • Who • Why	• Classify • Compare • Contrast • Demonstrate • Explain • Extend • Illustrate • Infer • Interpret • Outline • Relate • Rephrase • Show • Summarize • Translate	• Apply • Build • Choose • Construct • Develop • Experiment with • Identify • Interview • Make use of • Model • Organize • Plan • Select • Solve • Utilize	• Analyze • Assume • Categorize • Classify • Compare • Conclusion • Contrast • Discover • Dissect • Distinguish • Divide • Examine • Function • Inference • Inspect • List • Motive • Relationships • Simplify • Survey • Take part in • Test for • Theme	• Agree • Appraise • Assess • Award • Choose • Compare • Conclude • Criteria • Criticize • Decide • Deduct • Defend • Determine • Disprove • Estimate • Evaluate • Explain • Importance • Influence • Interpret • Judge • Justify • Mark • Measure • Opinion • Perceive • Prioritize • Prove • Rate • Recommend • Rule on • Select • Support • Value	• Adapt • Build • Change • Choose • Combine • Compile • Compose • Construct • Create • Delete • Design • Develop • Discuss • Elaborate • Estimate • Formulate • Happen • Imagine • Improve • Invent • Make up • Maximize • Minimize • Modify • Original • Originate • Plan • Predict • Propose • Solution • Solve • Suppose • Test • Theory

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504											
M.E. AUTOMOTIVE ELECTRONICS									Version: 1.0		
Courses of Study and Scheme of Assessment (Regulations 2023)									Date: 15.12.2025		
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER I											
THEORY											
1.	ME23MA102	Graph Theory and Optimization Techniques	FC	4	3	1	0	4	40	60	100
2.	UX23RM201	Research Methodology and IPR	RM	3	2	1	0	3	40	60	100
3.	ME23AE301	Automotive Engine Management Systems	PC	3	3	0	0	3	40	60	100
4.	ME23AE302	Automotive Sensors and Systems	PC	3	3	0	0	3	40	60	100
5.	ME23AE303	Electric Vehicles and Architectures	PC	3	3	0	0	3	40	60	100
6.	ME23AE304	Automotive Embedded Hardware Design	PC	3	3	0	0	3	40	60	100
PRACTICAL											
7.	ME23AE305	Automotive Embedded Hardware Design Laboratory	PC	4	0	0	4	2	60	40	100
8.	ME23AE306	Embedded C Programming Laboratory	PC	4	0	0	4	2	60	40	100
EMPLOYABILITY ENHANCEMENT											
9.	UX23PT801	Technical Seminar / Case Study Presentation	EEC	2	0	0	2	NC	100	-	100
		Total		**29**	**17**	**2**	**10**	**23**	**460**	**440**	**900**
SEMESTER II											
THEORY											
1.	ME23AE307	Automotive Electronic Control Units Development	PC	3	3	0	0	3	40	60	100
2.	ME23AE308	Testing and Simulations of Automotive ECUs	PC	3	3	0	0	3	40	60	100
3.	ME23AE309	AUTOSAR-Based ECUs Development	PC	3	3	0	0	3	40	60	100
4.	ME23AE4XX	Professional Elective –I	PE	3	3	0	0	3	40	60	100
5.	ME23AE4XX	Professional Elective –II	PE	3	3	0	0	3	40	60	100
6.	ME23XX5XX	Open Elective - I	OE	3	3	0	0	3	40	60	100
7.	UX23MC701	Universal Human Values and Ethics	MC	3	1	0	2	NC	100	-	100
8.	UX23AC7XX	Audit Course	AC	2	2	0	0	NC	100	-	100
PRACTICAL											
9.	ME23AE310	Testing and Simulations of Automotive ECUs Laboratory	PC	4	0	0	4	2	60	40	100
EMPLOYABILITY ENHANCEMENT											
10.	UX23PT802	Research Paper Review and Presentation	EEC	2	0	0	2	1	100	-	100
		Total		**29**	**21**	**0**	**8**	**21**	**600**	**400**	**1000**

w.e.f. 22/12/2025
Passed in BoS Meeting held on 06/12/2025
Approved in Academic Council Meeting held on 15/12/2025

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504												
M.E. AUTOMOTIVE ELECTRONICS												
Courses of Study and Scheme of Assessment (Regulations 2023)												
Sl. No.	**Course Code**	**Course Title**	**Periods / Week**						**Maximum Marks**			
			CAT	**CP**	**L**	**T**	**P**	**C**	**IA**	**ESE**	**Total**	
SEMESTER III												
THEORY												
1.	ME23AE311	Advanced Driver Assistance Systems	PC	3	3	0	0	3	40	60	100	
2.	ME23AE4XX	Professional Elective – III	PE	3	3	0	0	3	40	60	100	
3.	ME23AE4XX	Professional Elective – IV	PE	3	3	0	0	3	40	60	100	
4.	ME23XX5XX	Open Elective - II	OE	3	3	0	0	3	40	60	100	
PRACTICAL												
5.	ME23AE601	Project Work – Phase I	PW	12	0	0	12	6	60	40	100	
Total				**24**	**12**	**0**	**12**	**18**	**220**	**280**	**500**	
SEMESTER IV												
PRACTICAL												
1.	ME23AE602	Project Work – Phase II	PW	24	0	0	24	12	60	40	100	
Total				**24**	**0**	**0**	**24**	**12**	**60**	**40**	**100**	
				Total Number of Credits: 74								

SEMESTER-WISE CREDITS DISTRIBUTION

SUMMARY							
Sl. No.	**Course Category**	**Credits per Semester**				**Credits**	**Credit %**
		I	**II**	**III**	**IV**		
1.	FC	4	-	-	-	4	5
2.	RM	3	-	-	-	3	4
3.	PC	16	11	3	-	30	41
4.	PE	-	6	6	-	12	16
5.	OE	-	3	3	-	06	8
6.	PW	-	-	6	12	18	24
7.	MC/AC	-	0	-	-	0	0
8.	EEC	0	1	-	-	1	1
Total		**23**	**21**	**18**	**12**	**74**	**100**

NOMENCLATURE

CAT	Category of Course	**FC**	Foundation Courses	**MC/AC**	Mandatory Courses/ Audit Courses
CP	Contact Periods	**RM**	Research methodology and IPR courses	**EEC**	Employability Enhancement Courses
L	Lecture Hours	**PC**	Professional Core Courses	**IA**	Internal Assessment
T	Tutorial Hours	**PE**	Professional Elective Courses	**ESE**	Semester End Examination
P	Practical Hours	**OE**	Open Elective Courses	**PW**	Project Work Courses
C	Credits				

w.e.f. 22/12/2025
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Approved in Academic Council Meeting held on 15/12/2025

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KNOWLEDGE INSTITUTE OF TECHNOLOGY (AUTONOMOUS), SALEM - 637504												
M.E. AUTOMOTIVE ELECTRONICS												
Courses of Study and Scheme of Assessment (Regulations 2023)												
Sl. No.	Course Code	Course Title	Periods / Week						Maximum Marks		
			CAT	CP	L	T	P	C	IA	ESE	Total
SEMESTER III											
THEORY											
1.	ME23AE311	Advanced Driver Assistance Systems	PC	3	3	0	0	3	40	60	100
2.	ME23AE4XX	Professional Elective – III	PE	3	3	0	0	3	40	60	100
3.	ME23AE4XX	Professional Elective – IV	PE	3	3	0	0	3	40	60	100
4.	ME23XX5XX	Open Elective - II	OE	3	3	0	0	3	40	60	100
PRACTICAL											
5.	ME23AE601	Project Work – Phase I	PW	12	0	0	12	6	60	40	100
		Total		**24**	**12**	**0**	**12**	**18**	**220**	**280**	**500**
SEMESTER IV											
PRACTICAL											
1.	ME23AE602	Project Work – Phase II	PW	24	0	0	24	12	60	40	100
		Total		**24**	**0**	**0**	**24**	**12**	**60**	**40**	**100**
				Total Number of Credits: 74							

SEMESTER-WISE CREDITS DISTRIBUTION

SUMMARY							
Sl. No.	Course Category	Credits per Semester				Credits	Credit %
		I	II	III	IV		
1.	FC	4	-	-	-	4	5
2.	RM	3	-	-	-	3	4
3.	PC	16	11	3	-	30	41
4.	PE	-	6	6	-	12	16
5.	OE	-	3	3	-	06	8
6.	PW	-	-	6	12	18	24
7.	MC/AC	-	0	-	-	0	0
8.	EEC	0	1	-	-	1	1
Total		**23**	**21**	**18**	**12**	**74**	**100**

NOMENCLATURE

CAT	Category of Course	**FC**	Foundation Courses	**MC/AC**	Mandatory Courses/ Audit Courses
CP	Contact Periods	**RM**	Research methodology and IPR courses	**EEC**	Employability Enhancement Courses
L	Lecture Hours	**PC**	Professional Core Courses	**IA**	Internal Assessment
T	Tutorial Hours	**PE**	Professional Elective Courses	**ESE**	Semester End Examination
P	Practical Hours	**OE**	Open Elective Courses	**PW**	Project Work Courses
C	Credits				

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PROFESSIONAL ELECTIVES												
SEMESTER II & III (Professional Electives – I to IV)												
VERTICAL – I (Embedded Systems and ECU Design)												
Sl. No.	**Course Code**	**Course Title**	**Periods / Week**						**Maximum Marks**			
			CAT	**CP**	**L**	**T**	**P**	**C**	**IA**	**ESE**	**Total**	
1.	ME23AE401	RTOS in Multi-Core Environment	PE	3	3	0	0	3	40	60	100	
2.	ME23AE402	Parallel Programming using Multi-Cores and GPUs	PE	3	3	0	0	3	40	60	100	
3.	ME23AE403	Open Source Hardware and Software System Design	PE	3	3	0	0	3	40	60	100	
4.	ME23AE404	Automotive EMI and EMC Standards	PE	3	3	0	0	3	40	60	100	
5.	ME2AE3405	Automotive Power Electronics and Motor Drives	PE	3	3	0	0	3	40	60	100	
6.	ME23AE406	Alternative Drives, Traction and Controls	PE	3	3	0	0	3	40	60	100	
7.	ME23AE407	AUTOSAR and ISO Standards for Automotive Systems	PE	3	3	0	0	3	40	60	100	
8.	ME23AE408	Vehicle Electrification and Powertrain Integration	PE	3	3	0	0	3	40	60	100	
VERTICAL – II (Automotive Communication, Networking and Diagnostics)												
9.	ME23AE409	Vehicular Information and Communication Systems	PE	3	3	0	0	3	40	60	100	
10.	ME23AE410	Automotive Fault Diagnostics	PE	3	3	0	0	3	40	60	100	
11.	ME23AE411	Automotive Telematics and IoT Connectivity	PE	3	3	0	0	3	40	60	100	
12.	ME23AE412	Data Acquisition and Signal Conditioning	PE	3	3	0	0	3	40	60	100	
13.	ME23AE413	Machine Vision System for Automotive	PE	3	3	0	0	3	40	60	100	
14.	ME23AE414	Emission Control and Diagnosis	PE	3	3	0	0	3	40	60	100	
15.	ME23AE415	Vehicle Safety Systems	PE	3	3	0	0	3	40	60	100	
16.	ME23AE416	Human Factors and Ergonomics in Automotive Design	PE	3	3	0	0	3	40	60	100	

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Sl. No.	Course Code	Course Title	Offered by	Periods / Week						Maximum Marks		
				CAT	CP	L	T	P	C	IA	ESE	Total
OPEN ELECTIVES												
Except M.E. Industrial Safety Engineering												
1.	ME23IS501	Environmental Safety	MECH	OE	3	3	0	0	3	40	60	100
2.	ME23IS502	Electrical safety	MECH	OE	3	3	0	0	3	40	60	100
3.	ME23IS503	Safety in Engineering Industry	MECH	OE	3	3	0	0	3	40	60	100
4.	ME23IS504	Design of Experiments	MECH	OE	3	3	0	0	3	40	60	100
5.	ME23IS505	Circular Economy	MECH	OE	3	3	0	0	3	40	60	100
Except M.E. Automotive Electronics												
6.	ME23AE501	Fundamentals of Automotive Electronics	ECE	OE	3	3	0	0	3	40	60	100
7.	ME23AE502	Vehicle Sensors and Electronic Control Systems	ECE	OE	3	3	0	0	3	40	60	100
8.	ME23AE503	Automotive Embedded Systems	ECE	OE	3	3	0	0	3	40	60	100
9.	ME23AE504	Automotive Communication Networks	ECE	OE	3	3	0	0	3	40	60	100
10.	ME23AE505	Introduction to Electric and Hybrid Vehicle Electronics	ECE	OE	3	3	0	0	3	40	60	100
Except M.E. Power Electronics and Drives												
11.	ME23PE501	Electric Vehicle Technology	EEE	OE	3	3	0	0	3	40	60	100
12.	ME23PE502	Energy Conservation and Management	EEE	OE	3	3	0	0	3	40	60	100
13.	ME23PE503	Smart Grid	EEE	OE	3	3	0	0	3	40	60	100
14.	ME23PE504	Solar PV and Energy Storage Systems	EEE	OE	3	3	0	0	3	40	60	100
Except M.E. Software Engineering												
15.	ME23SE501	Blockchain Technologies	CSE	OE	3	3	0	0	3	40	60	100
16.	ME23SE502	Deep Learning	CSE	OE	3	3	0	0	3	40	60	100
17.	ME23SE503	Full Stack Web Application Development	CSE	OE	3	3	0	0	3	40	60	100
18.	ME23SE504	Principles of Multimedia	CSE	OE	3	3	0	0	3	40	60	100
19.	ME23SE505	Cyber Physical Systems	CSE	OE	3	3	0	0	3	40	60	100
AUDIT COURSES												
1.	UX23AC701	Industrial Safety Engineering	MECH	AC	2	2	0	0	0	100	-	100
2.	UX23AC702	Sustainability Engineering	MECH	AC	2	2	0	0	0	100	-	100
3.	UX23AC703	Energy Audit	EEE	AC	2	2	0	0	0	100	-	100
4.	UX23AC704	Design Thinking	CSE	AC	2	2	0	0	0	100	-	100
5.	UX23AC705	Business Analytics	CSE	AC	2	2	0	0	0	100	-	100

w.e.f. 22/12/2025
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ME23MA102	GRAPH THEORY AND OPTIMIZATION TECHNIQUES	CP	L	T	P	C
		4	3	1	0	4
Programme & Branch	M.E- AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:	
1.	To introduce graph as mathematical model to solve connectivity related problems.
2.	To introduce fundamental graph algorithms.
3.	To familiarize the students with the formulation and construction of a Mathematical modelfor a linear programming problem in real life situation.
4.	To provide knowledge and training using non-linear programming under Limited resourcesfor the engineering and business problems.
5.	To understand the applications of simulation modelling in engineering problems.

UNIT-I	GRAPHS	9+3
	Graphs and graph models - Graph terminology and special types of graphs - Matrix representation of graphs and graph isomorphism - Connectivity - Euler and Hamilton paths - Problems.	
UNIT-II	GRAPH ALGORITHM	9+3
	Graph Algorithms - Directed graphs - Some basic algorithms- Shortest path algorithms - Depth - First search on a graph - Theoretic algorithms - Performance of graph theoretic algorithms - Graph theoretic computer languages.	
UNIT-III	LINEAR PROGRAMMING	9+3
	Formulation - Graphical solution - Simplex method - Two-phase method -Transportation and Assignment Models.	
UNIT - IV	NON-LINEAR PROGRAMMING	9+3
	Constrained Problems - Equality constraints - Lagrangean Method - Inequality constraints - Karush - Kuhn-Tucker (KKT) conditions- Quadratic Programming.	
UNIT-V	SIMULATION MODELLING	9+3
	Monte Carlo Simulation - Types of Simulation - Elements of Discrete Event Simulation - Generation of Random Numbers - Applications to Queuing systems.	
	Total (LT) : 60 Periods	
	Course Outcomes: Upon completion of this course, the students will be able to:	**BLOOM'S Taxonomy**
CO1	Apply graph ideas is solving connectivity related problems.	L3 - Apply
CO2	Apply fundamental graph algorithms to solve certain optimization problems.	L3 - Apply
CO3	Formulate and construct mathematical models for linear programming problems and solve thetransportation and assignment problems.	L3 - Apply

w.e.f. 13/10/2025
Passed in BoS Meeting held on 18/09/2025
Approved in Academic Council Meeting held on 11/10/2025

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CO4	Model various real life situations as optimization problems and effect their solution throughNon-linear programming.	L3 – Apply
CO5	Apply simulation modeling techniques to problems drawn from industry management andother engineering fields.	L3 – Apply
	REFERENCE BOOKS:	
1.	Taha H.A, "Operation Research: An Introduction", Ninth Edition, Pearson Education, New Delhi, 2010.	
2.	Gupta P. K, and Hira D.S., "Operation Research", Revise Edition, S. Chand and Company Ltd., 2012.	
3.	Sharma J.K., "Operation Research", 3rd Edition, Macmillan Publishers India Ltd., 2009.	

Mapping of COs with POs

COs	POs					
	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	0	1	1	0	0
CO2	2	0	1	1	0	0
CO3	2	0	1	1	0	0
CO4	2	0	1	1	0	0
CO5	2	0	1	1	0	0
Average	**2.0**	**0**	**1.0**	**1.0**	**0**	**0**

1-Low, 2-Medium, 3-High.

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UX23RM201	RESEARCH METHODOLOGY AND IPR	CP	L	T	P	C
		3	2	1	0	3
Programme & Branch	Common to all M.E (SE,AE,PED,ISE) and M.C.A	Version: 1.0				

Course Objectives:		
1.	To analyze the significance of research and formulate well-defined research questions.	
2.	To apply appropriate research methods and critically evaluate research articles.	
3.	To create well-structured research papers and utilize research tools proficiently.	
4.	To produce effective technical reports and deliver impactful presentations.	
5.	To understand forms of intellectual property and analyze their implications on technological research and international cooperation.	
UNIT–I	CONCEPT OF RESEARCH	6+3
	Meaning and Significance of Research -Skills, Habits and Attitudes for Research -Time Management -Status of Research in India -Why, How, and What a Research is? -Types and Process of Research -Outcome of Research -Sources of Research Problem -Characteristics of a Good Research Problem -Errors in Selecting a Research Problem -Importance of Keywords -Literature Collection -Analysis -Citation Study - Gap Analysis -Problem Formulation Techniques.	
UNIT–II	RESEARCH METHODS AND JOURNALS	6+3
	Interdisciplinary Research -Need for Experimental Investigations -Data Collection Methods -Appropriate Choice of Algorithms / Methodologies / Methods -Measurement and Result Analysis -Investigation of Solutions for Research Problem -Interpretation -Research Limitations -Journals in Science/Engineering -Indexing and Impact factor of Journals -Citations- h Index - i10 Index -Journal Policies -How to Read a Published Paper -Ethical Issues Related to Publishing- Plagiarism and Self-Plagiarism.	
UNIT–III	PAPER WRITING AND RESEARCH TOOLS	6+3
	Types of Research Papers - Original Article/Review Paper/Short Communication/Case Study-When and Where to Publish? - Journal Selection Methods -Layout of a Research Paper -Guidelines for Submitting the Research Paper -Review Process - Addressing Reviewer Comments -Use of tools / Techniques for Research -Hands-on Training related to Reference Management Software -EndNote - Introduction to Origin, SPSS, etc. -Software for Detection of Plagiarism.	
UNIT – IV	EFFECTIVE TECHNICAL THESIS WRITING/PRESENTATION	6+3
	How to Write a Report - Language and Style -Format of Project Report - Use of Quotations -Method of Transcription Special Elements -Title Page - Abstract - Table of Contents - Headings and	

w.e.f. 13/10//2025
Passed in BoS Meeting held on 09/05/2025
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	Sub-Headings -Footnotes - Tables and Figures - Appendix - Bibliography etc. -Different Reference Formats -Presentation using PPTs.	
UNIT – V	**NATURE OF INTELLECTUAL PROPERTY**	**6+3**
	Patents - Designs - Trade and Copyright - Process of Patenting and Development - Technological research - innovation - patenting - Development International Scenario - International Cooperation on Intellectual Property - Procedure for Grants of Patents.	
	Total(LT)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate the importance and objectives of research in contributing to knowledge and solving real-world problems.	L2 - Understand
CO2	Experiment with data collection techniques, choosing fitting approaches to ensure sound research framework and methodology.	L3 - Apply
CO3	Utilize research & analytic tools for enhancing the research publication	L2 - Understand
CO4	Apply knowledge to produce presentations and technical reports that effectively communicate research findings.	L3 - Apply
CO5	Explain types of intellectual property and comprehend patenting as essential for safeguarding innovation and creativity.	L2 - Understand
	REFERENCE BOOKS:	
1.	Cooper, Donald R.; Schindler, Pamela S.; Sharma, J. K., "Business Research Methods", 12th Edition, Tata McGraw Hill Education, 2012.	
2.	DePoy, Elizabeth; Gitlin, Laura N., "Introduction to Research: Understanding and Applying Multiple Strategies", 6th Edition, Elsevier Health Sciences, 2015.	
3.	Walliman, Nicholas, "Research Methods: The Basics", 3rd Edition, Routledge, 2017.	
4.	Bettig, Ronald V., "Copyrighting Culture: The Political Economy of Intellectual Property", 1st Edition, Routledge, 2018.	
5.	The Institute of Company Secretaries of India, Statutory body under an Act of parliament, "Professional Programme Intellectual Property Rights, Law and practice", September 2013.	

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	3				
CO2	3	3				
CO3	3	3				
CO4	3	3			1	
CO5	3	3			1	
Avg.	**3.0**	**3.0**			**1.0**	
1-Low,2-Medium,3-High						

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ME23AE301	AUTOMOTIVE ENGINE MANAGEMENT SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				
Course Objectives:						
1.	To understand the fundamentals of automotive systems and the role of Electronic Control Units (ECUs) in modern vehicles.					
2.	To study the operation and control of petrol engine management systems.					
3.	To learn the concepts and evolution of diesel engine management systems.					
4.	To explore powertrain systems, including transmission types and hybrid powertrain control strategies.					
5.	To understand suspension, steering, and braking systems and their impact on vehicle performance and safety.					
UNIT–I	**INTRODUCTION TO AUTOMOTIVE SYSTEMS**	**9**				
	Electronic Control Unit(ECU) Design: The concepts of ECU design for automotive applications, Need for ECUs, advances in ECUs for automotive, design complexities of ECUs, V-Model for Automotive ECU's Architecture, analog, and digital interfaces. Basics of Engine Control systems: IC engines operation – Petrol and Diesel- IC engine as a propulsion source for Automobiles - need for engine controls and management - Control objectives - fuel efficiency - emission limits and vehicle performance- advantages of using electronic engine controls.					
UNIT–II	**Petrol Engine Management Systems**	**9**				
	Basics of Petrol engine controls: Electronic ignition - multi-point fuel injection - direct injection - Basics of ignition system and fuel injection system- Architecture of an EMS with multi point fuel injection.					
UNIT–III	**Diesel Engine Management Systems**	**9**				
	Basics of Diesel engine Controls: Evolution of diesel engine controls- in-line fuel pump - rotary fuel pump- EGR control - Electric motor driven fuel pump - electronic fuel injection control and timing.					
UNIT–IV	**POWERTRAIN SYSTEMS**	**9**				
	Transmission Systems: Introduction to manual- automatic- CVT and DCT transmissions. Hybrid Powertrains: Core concepts of integrating electric motors with combustion engines. Powertrain Control: Control strategies for engine and transmission management.					
UNIT– V	**VEHICLE SUSPENSION- STEERING- AND BRAKING**	**9**				
	Suspension Systems: Basic components and types of suspension (e.g.- independent- solid axle). Steering Mechanisms: Overview of steering systems- power-assisted steering. Braking Systems: Principles of disc and drum brakes- anti-lock braking (ABS) systems.					

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	Vehicle Handling: How suspension and steering systems affect handling and stability.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain different vehicle types (ICE- EVs- hybrids) and their operating principles.	L2 - Understand
CO2	Explain the working principles of internal combustion engines- hybrid powertrains and their control mechanisms.	L2 - Understand
CO3	Identify various types of suspension systems and their impact on vehicle handling and stability.	L3 - Apply
CO4	Illustrate the interaction between sensors, actuators, and ECUs in modern vehicles with reference to diagnostic and OBD-II functionalities.	L3 - Apply
CO5	Assess the impact of safety technologies such as airbags- seat belts- and crash safety standards.	L3 - Apply
	REFERENCE BOOKS:	
1.	David Crolla, "Automotive Engineering: Powertrain, Chassis System and Vehicle Body", Butterworth-Heinemann Ltd, 2016.	
2.	Jack Erjavec, Rob Thompson, "Automotive Technology: A Systems Approach", Cengage Learning, 2013.	
3.	John B. Heywood, "Internal Combustion Engine Fundamentals", McGraw-Hill Education, 1988.	
4.	Tom Denton, "Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives", Routledge, 2014.	
5.	Tom Denton, "Automotive Electrical and Electronic Systems", Routledge, 2015.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1	2	1		
CO2	1		2	1		
CO3	1		2	1	2	
CO4	1		2	1	2	
CO5	1		2	1	2	
Avg.	**1.0**	**1.0**	**2.0**	**1.0**	**2.0**	

1–Low- 2 –Medium- 3–High.

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ME23AE302	AUTOMOTIVE SENSORS AND SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	**M.E. AUTOMOTIVE ELECTRONICS**	**Version: 1.0**				

Course Objectives:	
1.	To understand the Integration of Electronics in Automotive.
2.	To explore Sensors- Actuators- and Communication Protocols.
3.	To analyze Signal Processing and Converters.
4.	To understand in Automotive Radar and LIDAR Systems.
5.	To study Advanced Driver Assistance Systems.

UNIT–I	**AUTOMOTIVE SENSORS**	**9**
	Automotive Sensors: Vehicle Body: - Torque sensors/ Force sensors- Sensors Flap air flow sensors- Temperature sensor-Ultrasonic sensors- Ranging radar (ACC) Power Train: - Fuel level sensors- Speed and RPM sensors- Lambda Oxygen sensor- Hotwire air mass meter Chassis: - Steering wheel angle sensor- Vibration and acceleration sensors- Pressure sensors- Speed and RPM sensors.	
UNIT–II	**BODY ELECTRONICS AND SAFETY**	**9**
	Body Electronics Applications: Automotive alarms- lighting systems- central locking- electric windows. Active and passive safety: Anti-lock Braking (ABS)-Electronic Electronic Brakeforce Distribution- Electronic Stability Program (ESP)- Hill Hold Control- adaptive cruise control- traction control- and active suspension systems.	
UNIT–III	**SIGNAL PROCESSING AND CONVERTERS**	**9**
	Introduction to Signal Processing: Discrete-time signals and systems- Time-domain analysis and introduction to digital filters. Conversion Techniques: Analog-to-Digital Converters (ADCs) and Digital-to-Analog Converters (DACs) - Sampling and reconstruction processes. Automotive Networking Protocols: Overview of LIN- CAN- FlexRay- and MOST protocols for vehicle communication.	
UNIT– IV	**AUTOMOTIVE RADAR AND LIDAR**	**9**
	Automotive Radar Systems: Frequency band allocation and international standards. Components: Antennas- radio frequency front end- signal processors. Waveform generation and range estimation. LIDAR Technology in Vehicles: Principles and applications of LIDAR in automotive systems. Vehicle Communication Systems: Digital engine control systems- Automotive controllers and introduction to On-Board Diagnostics (OBD)- Basics of Battery Management Systems (BMS).	
UNIT – V	**ADVANCED DRIVER ASSISTANCE SYSTEMS (ADAS) AND AUTONOMOUS VEHICLES**	**9**
	Adaptive cruise control- lane departure warning- blind-spot monitoring- Parking assistance and collision avoidance systems.	
	Total (L)	**45 Periods**

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	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes:Upon completion of this course the students will be able to:	**BLOOM'S Taxonomy**
CO1	Describe the roles of body electronics, powertrain control, and chassis systems in modern vehicles.	L2 - Understand
CO2	Apply the use of digital signal processing techniques, ADCs, and DACs in converting and analyzing automotive signals.	L3 - Apply
CO3	Summarize the working principles of CAN, LIN, and FlexRay protocols in automotive networking.	L3 - Apply
CO4	Apply the concepts of radar and LIDAR sensing to explain their role in vehicle safety and automation systems	L3 - Apply
CO5	Illustrate the application of power electronics concepts in developing energy,efficient conversion systems for electric vehicles.	L3 - Apply

	REFERENCE BOOKS:
1.	Bosch, "Automotive Electrics and Automotive Electronics. System and components, Networking and Hybrid drive", Sixth edition, Springer, 2021.
2.	Najamuz Zaman, "Automotive Electronics Design Fundamental" first edition, Springer 2015.
3.	William B. Ribbens, "Understanding Automotive Electronics" Eighth Edition, Butterworth , Heinemann,2017.
4.	Emmanuel C. Ifeachor and Barrie W. Jervis , "Digital Signal Processing: A Computer,Based Approach", 4th Edition, 2018.

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1	2	1		2
CO2	1		2	1		2
CO3	1		2	1	2	2
CO4	1		2	1	2	2
CO5	1		2	1	2	2
Avg.	**1.0**	**1.0**	**2.0**	**1.0**	**2.0**	**2.0**

1-Low, 2-Medium, 3-High.

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ME23AE303	ELECTRIC VEHICLES AND ARCHITECTURES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To introduce standards- impacts and economy of electric vehicles.	
2.	To impart knowledge of electric and hybrid electric vehicle architectures.	
3.	To introduce battery testing- maintenance- and monitoring techniques.	
4.	To introduce EV charging infrastructure, standards, grid integration, and regulatory considerations.	
5.	To understand global EV market trends, assess the impact of policies and legislation.	
UNIT–I	**INTRODUCTION TO ELECTRIC VEHICLES**	**9**
	Electric vehicles (EVs) - advantages and impacts - EV market and promotion - Infrastructure - Legislation and regulation - Standardization - Energy efficiency - Assessing economy of EVs - Fuel economy - Fuel consumption - Greenhouse gas emissions.	
UNIT–II	**HYBRID VEHICLE ARCHITECTURES AND POWER MANAGEMENT**	**9**
	Electrical subsystems - Components of a hybrid vehicle - 48V systems - Hybrid vehicle architectures - Design of a Hybrid Electric Vehicle (HEV) -EV architecture - Power converters and motor control - Case studies.	
UNIT–III	**ENERGY STORAGE SYSTEMS IN ELECTRIC AND HYBRID VEHICLES**	**9**
	Energy storage systems - Batteries and battery parameters - Types and characteristics of EV batteries - Battery testing - Charging schemes - Battery monitoring - Battery load levelling - Energy management strategies – EV Chargers - On-board and off-board chargers.	
UNIT–IV	**CHARGING INFRASTRUCTURE**	**9**
	EV Charging Infrastructure - Types of chargers (Level 1- Level 2- DC fast charging)- Charging Stations and Grid Integration- Electric Vehicle Supply Equipment: Standards and regulations for charging equipment.	
UNIT–V	**EV MARKET INTEGRATION**	**9**
	EV Market Trends and Future Growth- Impact of Legislation on EV Growth- Environmental Impact- Case Studies and Future Trends.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**

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CO1	Understand the impact of Electric Vehicles.	L2 – Understand
CO2	Analyze the fundamentals of Electric and Hybrid Vehicle technologies.	L4 – Analyse
CO3	Design and develop energy storage systems for electric vehicles.	L3 – Apply
CO4	Apply charging infrastructure concepts.	L3 – Apply
CO5	Analysis the environmental impact of electric vehicles.	L4 - Analyse
	REFERENCE BOOKS:	
1.	John G. Hayes, G. Abas Goodarzi, "Electric Powertrain Energy Systems, Power Electronics and Drives for Hybrid, Electric and Fuel Cell Vehicles", John Wiley and Sons, 2018.	
2.	Tom Denton, "Electric and Hybrid Vehicles", Routledge, 2016.	
3.	James Larminie, John Lowry, "Electric Vehicle Technology Explained", 2nd Edition, John Wiley and Sons, 2012.	
4.	Sheldon S. Williamson, "Energy Management Strategies for Electric and Plug", in Hybrid Electric Vehicles, Springer-Verlag New York- 2013.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1		1	1	1	
CO2	1		1	1	1	
CO3	1		1	1	1	
CO4	1		1	1	2	
CO5	1		1	1	1	
Avg.	**1.0**		**1.0**	**1.0**	**1.0**	

1-Low- 2 –Medium- 3–High.

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ME23AE304	AUTOMOTIVE EMBEDDED HARDWARE DESIGN	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To understand the features, architecture, and tools used for STM32 microcontroller development.	
2.	To learn how to configure and use GPIOs for basic interfacing and communication.	
3.	To study timers and PWM functions for control and timing applications.	
4.	To explore ADC configuration and signal conversion in STM32.	
5.	To apply CAN communication for data exchange in embedded and automotive systems.	
UNIT–I	**STM32 MICROCONTROLLERS**	**9**
	Overview of STM32 family and features- Development tools: TM32CubeMX- STM32CubeIDE- Setting up the development environment- Development tools: STM32CubeMX- STM32CubeIDE- Keil MDK- IAR - Embedded Workbench- Introduction to ARM Cortex-M core- Register architecture and memory organization.	
UNIT–II	**GPIO AND PERIPHERAL INTERFACING**	**9**
	Configuring GPIOs: Input and Output modes- Interfacing LEDs and push buttons- UART communication and data transmission.	
UNIT– III	**TIMERS AND PWM**	**9**
	Overview of Timer Peripherals- Timer Modes- Time Delays and Periodic Interrupts- Prescaler and Auto-Reload Registers- PWM Principles- PWM in STM32 and applications.	
UNIT – IV	**ANALOG-TO-DIGITAL CONVERSION (ADC)**	**9**
	Overview of ADC in STM32- Setting Up ADC in STM32: Selecting the ADC Channel- Configuring ADC Resolution- Sampling Time Configuration- ADC Conversion Mode.	
UNIT–V	**CONFIGURING CAN ON STM32**	**9**
	STM32 microcontroller features and architecture for ECU applications- CAN architecture in STM32 microcontrollers- CAN peripheral initialization using STM32CubeMX- Understanding CAN filters- baud rates- and identifiers.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**

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CO1	Configure GPIO pins for input and output operations.	L3 - Apply
CO2	Configure and utilize timers for real-time applications and implement Pulse Width Modulation (PWM).	L3 - Apply
CO3	Interface analog sensors to the microcontroller for data acquisition.	L3 - Apply
CO4	Select appropriate ADC channels, configure resolution, and configure sampling time to match application requirements.	L4 – Analyse
CO5	Use STM32CubeMX to initialize and configure the CAN peripheral for automotive communication systems.	L3 -Apply
	REFERENCE BOOKS:	
1.	M. P. Singh, "STM32 Embedded Systems Programming: With STM32CubeIDE, STM32CubeMX, and HAL Library", Springer, 2020.	
2.	Rajeev K. Dube, "Automotive Embedded Systems Handbook", CRC Press, 2019.	
3.	Jonathan Valvano, "Embedded Systems: Real-Time Operating Systems for ARM Cortex M Microcontrollers", CreateSpace Independent Publishing Platform, 2015.	
4.	Dr. Majid Maleki, "STM32 Microcontroller and Embedded Systems: Using Arduino IDE", CreateSpace Independent Publishing Platform, 2019.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2		2	2	2	
CO2	2		2	2	2	
CO3	1		2	2	2	
CO4	1		2	3	2	
CO5	2		2	2	2	
Avg.	**2.0**		**2.0**	**2.0**	**2.0**	
1–Low- 2 –Medium- 3–High.						

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ME23AE305	AUTOMOTIVE EMBEDDED HARDWARE DESIGN LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:	
1.	To understand the STM32 Microcontroller Architecture and Tools
2.	To develop GPIO and Peripheral Interfacing Skills
3.	To implement Timers- PWM- and ADC Applications
4.	To learn Communication Protocols
5.	To design and Debug Embedded Systems
LIST OF EXPERIMENTS	
1.	Setting Up STM32 Development Environment • Install STM32CubeMX and STM32CubeIDE • Set up a simple project for an STM32 microcontroller • Configure basic GPIO pins to blink an LED • Debug and run the project
2.	GPIO and Digital I/O Interface • Configure digital input pins for switches and output pins for LEDs • Implement a simple push-button interface to control LED state • Debug the system and verify the operation of digital I/O
3.	Analog Input and Output Interface (ADC/DAC) • Interface a potentiometer (variable resistor) to the ADC input • Read the sensor value and display it on a serial monitor • Implement a DAC output to control an external analog device.
4.	Pulse Width Modulation (PWM) for Motor Control • Configure a PWM output on STM32 for motor control • Vary the duty cycle of PWM to change motor speed • Implement motor direction control using an H-Bridge circuit • Test motor speed and direction control using the STM32
5.	UART Communication (Serial Communication) • Set up UART communication between STM32 and a PC using a USB-to-UART converter • Implement sending and receiving messages via UART • Test bi-directional communication with a sensor
6.	CAN Protocol Communication • Configure CAN peripheral • Establish CAN communication for data transfer
7.	SPI Communication for Sensor Interface • Interface an SPI-based sensor (e.g.- temperature or pressure sensor) with STM32 • Set up SPI communication using STM32CubeMX • Read sensor data and display it via UART • Handle sensor data processing and display
8.	I2C Communication with External Devices • Interface an I2C EEPROM or temperature sensor to STM32 • Configure the I2C interface using STM32CubeMX • Read and write data to EEPROM using I2C • Display read data via UART
	Total (P) — 60 Periods

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	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO 1	Demonstrate the ability to configure GPIO pins and use basic debugging tools to run embedded applications.	L3 – Apply
CO 2	Ability to design and implement digital I/O interfaces, such as connecting and controlling LEDs with switches.	L3 – Apply
CO 3	Interface analog sensors with the STM32 ADC and read sensor data for real-time processing.	L3 – Apply
CO 4	Practical knowledge in adjusting PWM duty cycles to control motor behavior.	L3 – Apply
CO 5	Develop proficiency in configuring and using UART, SPI, I2C, and CAN communication protocols with STM32 microcontrollers.	L3 – Apply

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	1	2	2	3	2	2
CO2	1	2	2	3	2	2
CO3	1	2	2	3	2	2
CO4	1	2	2	3	2	2
CO5	1	2	2	3	2	2
Avg.	**1.0**	**2.0**	**2.0**	**3.0**	**2.0**	**2.0**
1–Low- 2 –Medium- 3–High.						

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ME23AE306	EMBEDDED C PROGRAMMING LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:	
1.	To learn Embedded C programming for controlling and interfacing hardware devices.
2.	To develop embedded applications that handle input/output devices and communication protocols.
3.	To interface and process analog and digital signals using ADC and PWM.
4.	To implement interrupt handling- LED control- and interfacing with sensors and displays.
5.	To establish reliable data communication between devices using UART and related protocols.

LIST OF EXPERIMENTS		
1.	Write an Embedded C program to blink an LED on and off at a specific interval.	
2.	Write an Embedded C program for Push-Button to Control LED.	
3.	Write an Embedded C program to vary PWM for LED Brightness Control.	
4.	Write an Embedded C program to Interface a Potentiometer with ADC.	
5.	Write an Embedded C program for Interfacing an External LCD Display.	
6.	Write an Embedded C program for Interrupt Programming.	
7.	Write an Embedded C program for UART Communication.	
	Total (P)	**60 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Configure and manipulate hardware peripherals (GPIO- ADC- PWM- UART- etc.) using Embedded C programming.	L3 – Apply
CO2	Implement control systems- data input/output- and communication protocols- enhancing the ability to develop embedded applications with practical real-world interactions.	L3 – Apply
CO3	Demonstrate proficiency in programming, debugging, and testing Embedded C code for real-time hardware interfacing and embedded system applications.	L3 – Apply
CO4	Build skills to manage interrupts- control LED brightness- interface with sensors and displays	L3 – Apply
CO5	Establish reliable communication using UART and other protocols.	L3 – Apply

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	1			3	2	
CO2	1			3	2	
CO3	1			3	2	
CO4	1			3	2	
CO5	1			3	2	
Avg.	**1.0**			**3.0**	**2.0**	
1-Low- 2 -Medium- 3-High.						

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UX23PT801	TECHNICAL SEMINAR / CASE STUDY PRESENTATION	CP	L	T	P	C
		2	0	0	2	NC
Programme & Branch	**Common to all M.E (SE,AE,PED,ISE)**	**Version: 1.0**				

Course Objectives:

1.	To encourage the students to study advanced engineering developments
2.	To prepare and present the technical and case study reports

Method of Evaluation:

The students need to identify an area of interest or topic in their programme of study or case study and prepare a 5-10 page report and a presentation. Based on the report and presentation, the course is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he has to make such presentation in the subsequent semesters. The evaluation guidelines will be issued by the Head of the Department before the commencements of the course. The objectives are improving literature searching capabilities, comprehension and ability to write reports and to make presentations. It is assessed in Internal Assessment mode only and no End Semester Examination.

	Total (P)	**30 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Perform the review and present technological developments in their field	L3 - Apply
CO2	Interpret the case study report and make a decision	L3 - Apply

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
1		3				
2		3				
Avg.		**3.0**				

1–Low, 2 –Medium, 3–High.

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ME23AE307	AUTOMOTIVE ELECTRONIC CONTROL UNITS DEVELOPMENT	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:	
1.	To understand what Automotive ECUs are and how they work in vehicles.
2.	To learn basic vehicle communication systems like CAN, LIN, and FlexRay using TSMaster.
3.	To know how ECUs are developed and tested using TSMaster tools.
4.	To use common ECU design and simulation tools for creating and checking ECU functions.
5.	To learn about advanced systems like ADAS, autonomous vehicles, and future V2X communication.

UNIT–I	**INTRODUCTION TO AUTOMOTIVE ECUS**	**9**
	Overview of Automotive ECUs Definition - role of ECUs in modern vehicles - Importance of ECUs for vehicle functionality – safety - performance - Types of ECUs in vehicles: Powertrain ECU - Body Control Module (BCM) – Infotainment - Safety ECUs - Communication protocols used in automotive systems: CAN – LIN – FlexRay – Ethernet - Automotive ECU Architecture Basic structure of an ECU: Microcontroller – sensors – actuators - communication interfaces - Power supply and signal conditioning in automotive environments – Automotive - grade components and their selection.	
UNIT–II	**AUTOMOTIVE COMMUNICATION PROTOCOLS**	**9**
	CAN (Controller Area Network) Communication - Overview of CAN protocol: Its role and importance in automotive communication - CAN message structure: Data frame - remote frame - error frame - overload frame - Integration of TSMaster with CAN for real - time data monitoring - logging and testing - Practical use case: Setting up CAN communication with TSMaster and monitoring traffic - Other Automotive Communication Protocols - LIN (Local Interconnect Network): Basic concepts and automotive use cases-FlexRay: High-speed communication for safety - critical ECUs.	
UNIT– III	**ECU DEVELOPMENT LIFE CYCLE WITH TSMaster**	**9**
	ECU Design Process Requirements gathering for automotive applications-Design and architecture of ECUs: Hardware and software co-design - Using TSMaster in the ECU design and validation process: Managing test plans - real-time monitoring - Testing and Validation Using TSMaster for Hardware-in-the-loop (HIL) testing - Software-in-the-loop (SIL) simulation - Setting up automated tests for ECU software using TSMaster - Managing test reports and error diagnostics with TSMaster.	
UNIT – IV	**ECU DEVELOPMENT TOOLS AND PLATFORMS WITH TSMaster**	**9**
	Development Tools for ECU Design - Overview of tools for ECU design and programming (e.g: NXP S32K, Infineon, STMicroelectronics tools) - TSMaster as a platform for ECU design and testing integration - Code generation tools (MATLAB/Simulink, Embedded Coder) - integration with TSMaster - Automotive ECU Simulation Platforms: Introduction to Hardware-in-the-loop (HIL) - Model-in-the-loop (MIL) simulation - Use	

	of TSMaster for HIL and MIL testing to validate ECU functionality and performance.	
UNIT–V	**ADVANCED AUTOMOTIVE ECUS**	**9**
	Advanced Driver - Assistance Systems (ADAS) - Autonomous Vehicles - Role of ECUs in ADAS systems: Radar, LiDAR, cameras, and their integration with ECUs - Integration of TSMaster with ADAS systems for performance testing and diagnostics - Use of TSMaster for autonomous vehicle systems monitoring.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the role of ECUs in modern vehicles and explain their architecture, functions, safety, and performance benefits.	L2 – Understand
CO2	Apply automotive communication protocols such as CAN, LIN, FlexRay, and Ethernet to design and test vehicle network systems.	L3 – Apply
CO3	Use tools like TSMaster to test and validate ECUs using HIL, SIL, and automated testing methods.	L3 – Apply
CO4	Demonstrate how ECUs are used in advanced applications like ADAS, autonomous vehicles, and V2X communication systems.	L3 – Apply
CO5	Use recent ECU technologies to model and illustrate autonomous vehicle functions	L3 – Apply
	REFERENCE BOOKS:	
1.	Rolf Isermann, "Automotive Control Systems for Engine, Driveline, and Vehicle", Springer, 2017.	
2.	William B. Ribbens, "Understanding Automotive Electronics", Elsevier, 8th Edition, 2017.	
3.	Hans-Jurgen Reuter, "Automotive Mechatronics: Automotive Networking, Driving Stability Systems", Electronics, Springer, 2013.	
4.	Joerg Schaeuffele and Thomas Zurawka, "Automotive Software Engineering: Principles, Processes, and Methods", SAE International, 2nd Edition, 2016.	

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	2	2	2	2	
CO2	2	2	2	2	2	
CO3	2	2	2	2	2	
CO4	2	2	2	2	2	1
CO5	2	2	2	2	2	1
Avg.	**2.0**	**2.0**	**2.0**	**2.0**	**2.0**	**1**

1-Low, 2 -Medium, 3-High.

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ME23AE308	TESTING AND SIMULATIONS OF AUTOMOTIVE ECUs	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:

1.	To understand model-based design and simulation methods like MIL and HIL used in ECU development.
2.	To use MATLAB/Simulink to build, simulate, and analyze automotive ECU models.
3.	To develop clear and effective test cases for validating automotive ECU functions and performance.
4.	To execute MIL/HIL tests and generate reports to evaluate and improve ECU behavior.
5.	To learn advanced simulation techniques for ADAS, autonomous systems, and real-time ECU validation.

UNIT–I	**INTRODUCTION TO MODEL-BASED DESIGN AND SIMULATION**	**9**
	Overview of Model-based Design: Introduction to model based design in automotive systems - Importance of simulation in the development of automotive ECUs - Benefits of using simulation techniques in the validation of ECUs - Model-in-the-loop (MIL) Simulation - Understanding MIL: Concept and role in ECU validation - Workflow of MIL simulation - MATLAB/Simulink environment for MIL: Model creation – simulation - analysis - Hardware-in-the-loop (HIL) Simulation - Introduction to HIL: Key differences from MIL - Setting up HIL testing for ECUs - Tools and platforms for HIL simulation (e.g. dSPACE, NI, and ETAS) - HIL simulation for real-time validation of ECU performance.	
UNIT–II	**MATLAB/SIMULINK FOR MODEL-BASED SIMULATION**	**9**
	Introduction to MATLAB/Simulink for Automotive Applications: Basics of MATLAB and Simulink - Introduction to the development environment-Simulink blocks for automotive simulation - Designing and testing simple automotive models - Simulating Automotive ECU Systems - Creating models for ECUs – Sensors – actuators - communication protocols - Simulating ECU functions such as control algorithms - data exchange – timing - Running simulations for functional and performance validation of automotive ECUs - Simulation for Functional Validation: Using Simulink to validate basic ECU functionalities (e.g., engine control, braking systems) - Simulating feedback loops and sensor - actuator dynamics - Monitoring signals and analyzing results from the simulation.	
UNIT– III	**TEST CASE DEVELOPMENT FOR AUTOMOTIVE ECUs**	**9**
	Writing Test Cases for ECU Validation: Introduction to test cases in automotive ECU development - Writing test cases for functional - integration - system testing of automotive ECUs - Test case structure - Inputs - expected outputs - conditions and constraints - Designing Test Cases for ECU Functions - Design test cases for various ECU functions - Communication protocols (CAN- LIN- Ethernet) - signal processing - control algorithms - safety features - Test case design for different testing levels (unit testing- system testing) - Test Case Design for Performance Validation - Performance testing of ECUs – Timing - resource usage - load testing - Design test cases to evaluate ECU	

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	performance under different conditions - Stress testing - boundary testing - fault injection for ECUs.	
UNIT – IV	**TEST EXECUTION STRATEGIES AND REPORTING**	**9**
	Test Execution Strategies: Methods for running and executing test cases on simulated models or hardware (HIL and MIL) - Managing test cycles and evaluating outcomes - Automated testing vs manual testing in ECU validation - Tools for managing large-scale test cases (e.g., Vector CANoe, NI VeriStand) - Test Reporting and Diagnostics: Generating test reports from simulation results - Techniques for analysing test results: Identifying failures – errors and performance bottlenecks - Using test reports for debugging and improving ECU design.	
UNIT–V	**ADVANCED SIMULATION TECHNIQUES**	**9**
	Advanced Automotive Simulation - Simulation of complex systems - ADAS - autonomous driving – safety - critical ECUs - Simulating communication protocols in multi-ECU systems - Use of real-time simulation for validation of advanced automotive ECUs - Integration of MIL, HIL, and Real-World Testing: Combining MIL and HIL simulations for comprehensive validation - Best practices for integrating simulation models with real - world testing and deployment - Future trends in automotive ECU validation: AI and machine learning in simulation.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply model-based design techniques using tools like MATLAB/Simulink for simulating and analysing automotive ECU functionalities	L3 – Apply
CO2	Develop and validate simulation models for functional and performance testing of automotive ECUs	L3 – Apply
CO3	Design and execute structured test cases for ECU systems, focusing on functional, integration, and performance aspects, including stress and fault injection testing.	L3 – Apply
CO4	Apply advanced simulation techniques to validate complex automotive systems	L3 – Apply
CO5	Use simulation results and test reports to detect failures and enhance ECU operation.	L3 – Apply
	REFERENCE BOOKS:	
1.	Steven F. Barrett and Daniel J. Pack, "Embedded Systems: Design and Applications", Morgan & Claypool, 2nd Edition, 2019.	
2.	Jorg Schauffele and Thomas Zurawka, "Automotive Software Engineering: Principles, Processes, and Methods", Springer, 2nd Edition, 2013.	
3.	Hermann Kopetz, "Real-Time Systems: Design Principles for Distributed Embedded Applications", Springer, 3rd Edition, 2011.	
4.	Kirsten Matheus and Thomas Königseder, "Automotive Ethernet: The Definitive	

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	Guide", Springer, 2nd Edition, 2021.

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2	2	2	3	2	2
CO2	2	2	2	2	2	2
CO3	1	2	2	2	2	2
CO4	1	2	2	3	2	2
CO5	2	2	2	2	3	2
Avg.	**1.6**	**2.0**	**2.0**	**2.4**	**2.2**	**2.0**
1-Low, 2 -Medium, 3-High.						

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ME23AE309	AUTOSAR-BASED ECUs DEVELOPMENT	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To understand AUTOSAR Fundamental.	
2.	To explore AUTOSAR Architecture.	
3.	To master AUTOSAR Configuration and Toolchains.	
4.	To develop AUTOSAR Software Components (SWCs).	
5.	To testing and validation of AUTOSAR ECUs.	
UNIT–I	**INTRODUCTION TO AUTOSAR**	9
	Introduction to AUTOSAR: Definition and objectives of AUTOSAR (Automotive Open System Architecture) - Benefits of AUTOSAR: Standardization – scalability – reusability - AUTOSAR layers: Basic Software (BSW) - Runtime Environment (RTE) - Application Layer - AUTOSAR Standards: Classic Platform (CP) vs Adaptive Platform (AP) - Overview of AUTOSAR Release versions -Understanding AUTOSAR methodology and its application in automotive industry.	
UNIT–II	**AUTOSAR ARCHITECTURE AND LAYERS**	9
	Basic Software Layer (BSW): Description of the Basic Software architecture-Core components: ECU Abstraction Layer (ECU-AL) - Microcontroller Abstraction Layer (MCAL) - and Services - Software components within BSW: Communication Stack - Memory Services-Diagnostics- and I/O Modules - Runtime Environment (RTE): Role of RTE in AUTOSAR architecture - RTE communication model: Synchronous and Asynchronous communication - Understanding the interaction between Application Layer and BSW via RTE - Application Layer: Structure and role of the Application Layer in AUTOSAR - Creating and configuring AUTOSAR software components (SWCs) - Role of Application Software Components (SWCs) in controlling ECU functionality.	
UNIT– III	**AUTOSAR CONFIGURATION AND TOOLS**	9
	AUTOSAR Configuration Process: Overview of the AUTOSAR configuration process - Configuration of ECUs using AUTOSAR tools - AUTOSAR Toolchain: Introduction to the AUTOSAR toolchain – Configuration - Integration and Validation tools -A UTOSAR tools: DaVinci Configurator - Vector CANoe ETAS ISOLAR, and others - Code generation tools for AUTOSAR systems: Integration with Matlab/Simulink and Embedded Coder.	
UNIT – IV	**DEVELOPMENT PROCESS AND AUTOSAR SOFTWARE COMPONENT (SWC) DESIGN**	9
	Software Component Development: Introduction to AUTOSAR Software Components (SWC) - Designing SWCs for different automotive applications: Communication - Sensor Control – Actuation - Communication between SWCs via RTE - AUTOSAR Methodology for SWC Development: Development and integration of AUTOSAR SWCs using Simulink/Matlab - Modelling SWCs in Simulink and auto - generating C code using Embedded Coder - Mapping software components to the underlying BSW architecture.	

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UNIT–V	**TESTING AND VALIDATION OF AUTOSAR ECUS**	**9**
	Hardware-in-the-Loop (HIL) and Software-in-the-Loop (SIL) Testing: Introduction to HIL and SIL testing for AUTOSAR based systems - Setting up HIL simulation for AUTOSAR ECUs - Use of AUTOSAR compliant testing tools like CANoe, dSPACE, and ETAS - Diagnostic and Error Handling in AUTOSAR: Implementing diagnostic services in AUTOSAR ECUs - Error handling and fault tolerance mechanisms - Testing and validating diagnostic communication using UDS (Unified Diagnostic Services).	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Describe the different layers of AUTOSAR, including Basic Software (BSW), Runtime Environment (RTE), and the Application Layer.	L2 – Understand
CO2	Design and set up RTE software so that application software communicates properly with the hardware layers.	L3 – Apply
CO3	Implement AUTOSAR-compliant configurations in automotive ECUs using appropriate toolchains.	L3 – Apply
CO4	Design AUTOSAR Software Components (SWCs) for different automotive applications.	L3 – Apply
CO5	Implement diagnostic and error-handling mechanisms in AUTOSAR systems.	L3 – Apply
	REFERENCE BOOKS:	
1.	Johannes H. Huber, Michael D. O Reilly, and Jorg Tiedemann, "AUTOSAR – The Standardized Software Architecture for Automotive Control Systems", 1st Edition, 2020.	
2.	Ralf Eberhardt, Uwe Kiencke, "Automotive Control Systems: For Engine, Driveline, and Vehicle", 2nd Edition, 2016.	
3.	Martin Torngren, Paul K. Kuo, "Automotive Software Architectures: An Introduction", Springer Nature, 2015.	
4.	Bosch, "Automotive Handbook", Wiley, 9th Edition, 2020.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2		2	3	3	1
CO2	2		2	2	3	1
CO3	1		2	2	3	1
CO4	1		2	3	2	1
CO5	2		2	2	1	1
Avg.	**1.6**		**2.0**	**2.4**	**2.4**	**1.0**

1–Low, 2 –Medium, 3–High.

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UX23MC701	UNIVERSAL HUMAN VALUES AND ETHICS	CP	L	T	P	C
		3	1	0	2	NC
Programme & Branch	COMMON TO M.E. (SE,AE,PED,ISE), M.C.A & MBA (IEV)	Version: 1.1				

Course Objectives:

1.	To understand the concept of Universal Human Values.
2.	To discuss theoretical and practical implications of UHV.
3.	To relate the use of harmony in the family and society.
4.	To classify the harmony in the nature methods.
5.	To construct effective human values in personal and professional in life.

UNIT–I	**INTRODUCTION TO VALUE EDUCATION**	**3+6**
	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Understanding Value Education - Sharing about Oneself - Self-exploration as the Process for Value Education - Continuous Happiness and Prosperity – the Basic Human Aspirations - Exploring Human Consciousness - Happiness and Prosperity – Current Scenario - Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance . * **Experiential Learning:** Practice sessions to discuss natural acceptance in human being.	
UNIT–II	**HARMONY IN THE HUMAN BEING**	**3+6**
	Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self and Body - The Body as an Instrument of the Self - Understanding Harmony in the Self - Exploring Sources of Imagination in the Self- Harmony of the Self with the Body - Programme to ensure self-regulation and Health - Exploring Harmony of Self with the Body . * **Experiential Learning:** Practice sessions to discuss the awareness of self, help in managing stress and anxiety.	
UNIT–III	**HARMONY IN THE FAMILY AND SOCIETY**	**3+6**
	Harmony in the Family – the Basic Unit of Human Interaction - 'Trust' – the Foundational Value in Relationship - Exploring the Feeling of Trust - 'Respect' – as the Right Evaluation - Exploring the Feeling of Respect - Other Feelings , Justice in Human-to-Human Relationship - Understanding Harmony in the Society - Vision for the Universal Human Order - Exploring Systems to fulfil Human Goal. * **Experiential Learning:** Case Studies on Family and societal issues.	

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UNIT – IV	**HARMONY IN THE NATURE/EXISTENCE**	**3+6**
	Understanding Harmony in the Nature – Interconnectedness , self-regulation and Mutual Fulfilment among the Four Orders of Nature - Exploring the Four Orders of Nature - Realizing Existence as Co-existence at All Levels - The Holistic Perception of Harmony in Existence - Exploring Co-existence in Existence. * **Experiential Learning:** Case Study to discuss human being as cause of imbalance in nature, pollution, depletion of resources and role of technology.	
UNIT – V	**IMPLICATIONS OF THE HOLISTIC UNDERSTANDING - A LOOK AT PROFESSIONAL ETHICS**	**3+6**
	Natural Acceptance of Human Values - Definitiveness of (Ethical) Human Conduct - Exploring Ethical Human Conduct - A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order - Competence in Professional Ethics - Exploring Humanistic Models in Education - Holistic Technologies, Production Systems and Management Models -Typical Case Studies - Strategies for Transition towards Value-based Life and Profession - Exploring Steps of Transition towards Universal Human Order. * **Experiential Learning:** Case studies of holistic technologies and Professional ethics.	
	Total (LP) : 45 Periods	
	*** Internal Evaluation only.**	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the class room teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Interpret the concepts of Universal Human Values.	L2 - Understand
CO2	Summarize both theoretical and practical implications of Universal Human Values.	L2 - Understand
CO3	Build the harmony in family and society.	L3 - Apply
CO4	Practice harmony in all human existence.	L3 - Apply
CO5	Relate human values in both personal and professional life.	L2 - Understand
	REFERENCE BOOKS:	
1.	R.R Gaur, R Sangal, G P Bagaria, A foundation course in Human Values and professional Ethics – Teachers Manual, Excel books, New Delhi	

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2.	B L Bajpai, 2004, Indian Ethos and Modern Management, New Royal Book Co., Lucknow, Reprinted 2008.	
3.	Frankl, Viktor E. Yes to Life In spite of Everything, Penguin Random House, London, 2019.	
4.	Van Zomeren, M., & Dovidio, J. F. The Oxford Handbook of the Human Essence (Eds.), New York Oxford University Press, 2018.	
5.	R R Gaur, R Asthana, G P Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.	
6.	A.N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004.	
S.No.	**WEB REFERENCES:**	
1.	https://www.researchgate.net/publication/382495858_Universal_Human_Value	
2.	https://fdp-si.aicte-india.org/UHVII.php	
	VIDEO REFERENCES:	

S.No.	Publisher	Website link	Type of Content
1.	Dr.Rajneesh Arora	https://www.youtube.com/watch?v=SdXlB0k7klQ&list=PLWDeKF97v9SP7wSlapZcQRrT7OH0ZIGC4&index=2	Video Lecture
2.	Mr.Umesh Jadhav	https://www.youtube.com/watch?v=_g8Hw2APx8Q	Video Lecture

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ME23AE310	TESTING AND SIMULATIONS OF AUTOMOTIVE ECUs LABORATORY	CP	L	T	P	C
		4	0	0	4	2
Programme & Branch	**M.E. AUTOMOTIVE ELECTRONICS**	**Version: 1.0**				

Course Objectives:

1.	To learn the MATLAB/Simulink interface and build a simple braking system model for simulation.
2.	To create an engine control model and use MIL simulation to validate ECU logic and outputs.
3.	To perform HIL testing by connecting a real ECU to a simulation platform and analysing real-time data.
4.	To write and manage test cases for an ECU function, including inputs, outputs, and documentation.
5.	To model an automotive ECU in Simulink and validate its functionality and performance using test cases.

LIST OF EXPERIMENTS

1.	Introduction to MATLAB/Simulink Environment Familiarize with the MATLAB/Simulink interface and basic tools for automotive simulation. Create a simple model of a vehicle's braking system using basic blocks (sensors, actuators). Run the simulation and visualize signal behaviour.
2.	Model-in-the-Loop (MIL) Simulation for ECU Validation Create a Simulink model for an engine control system with sensors, actuators, and control algorithms. Perform MIL simulation to validate the logic of the ECU. Use MATLAB to monitor and analyze system outputs such as engine speed and fuel level.
3.	Hardware-in-the-Loop (HIL) Simulation Connect a real ECU to a simulation system using a real-time platform (e.g., dSPACE or NI VeriStand). Simulate the ECU's environment (automotive sensors and actuators) in MATLAB/Simulink. Collect and analyse data from the ECU in real-time to validate its performance.
4.	Writing and Managing Test Cases for ECU Select an ECU function (e.g., vehicle lighting control). Write test cases for functional validation, such as light intensity based on external conditions. Define inputs, expected outputs, and constraints for each test case. Document test results and create a test execution plan.
5.	Functional and Performance Validation of ECUs Create a Simulink model for a specific automotive ECU (e.g., powertrain ECU). Develop and execute test cases for functionality and performance validation. Analyse the results using MATLAB and diagnose any errors or performance issues.
6.	Testing Communication Protocols (CAN, LIN) Set up CAN communication in Simulink to simulate data exchange between two ECUs. Develop test cases for message transmission and error handling. Monitor communication data using MATLAB and analyse pr[illegible]

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	Test ECU behaviour under different communication conditions.

	Total(P)	**60 Periods**
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Use MATLAB/Simulink to model and simulate basic automotive systems.	L3 - Apply
CO2	Develop and validate ECU control logic using Model-in-the-Loop (MIL) simulation.	L3 - Apply
CO3	Perform Hardware-in-the-Loop (HIL) testing by interfacing real ECUs with simulation platforms.	L3 - Apply
CO4	Design and execute test cases for functional and performance validation of automotive ECUs.	L3 - Apply
CO5	Analyse ECU communication using CAN and LIN protocols under different operating conditions.	L4 - Analyse

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	1			3	2	
CO2	1			3	2	
CO3	1			3	2	
CO4	1			3	2	
CO5	1			3	2	
Avg.	**1.0**			**3.0**	**2.0**	
1-Low, 2 -Medium, 3-High.						

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UX23PT802	RESEARCH PAPER REVIEW AND PRESENTATION	CP	L	T	P	C
		2	0	0	2	1
Programme & Branch	Common to all M.E (ISE, AE, PED, SE) and M.C.A.	Version: 1.0				

Course Objectives:

1.	To learn scientific paper reading and wiring skills.
2.	To learn the literature review and report wiring skills.
3.	To understand the research gap and formulation of the research problem.

The work involves the following steps:

1. Assigning the faculty supervisor
2. Selecting a subject, narrowing the subject into a topic
3. Stating an objective.
4. Collecting the relevant bibliography (atleast 20 research papers)
5. Studying the papers understanding the authors contributions and critically analyzing each paper.
6. Preparing a 20-25 page literature review report
7. Preparing conclusions based on the literature review report.
8. Writing the Final Review Paper
9. Final Presentation to the review committee

Evaluation method:

A faculty supervisor will be assigned to each student. The supervisor will assign a topic to the student. The student has to review the literature pertaining to the topic, prepare a 20-25 page report and make a presentation. Minimum 20 research papers have to be reviewed out of which 60% have to be in the recent 05 years. The format for the research paper report and guidelines for assessment will be issued by the Head of the Department before the commencement of the course. The evaluation will be carried out based on the research paper report and presentation, and is evaluated for 100 marks. Minimum 50 marks is essential to pass. In case a student fails, he or she has to redo the course in the forthcoming semesters. Assessment is by Internal Assessment mode only no End Semester Examination.

	Total(P)	30 Periods
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Write a scientific review paper in their field	L3 - Apply
CO2	Identify the research gap and formulate the research problem	L3 - Apply

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1		3				
CO2		3				
Avg.		**3.0**				
1-Low, 2 -Medium, 3-High.						

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ME23AE311	ADVANCED DRIVER ASSISTANCE SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:	
1.	To learn ADAS basics, autonomy levels, key sensors, and how system components integrate.
2.	To understand camera, radar, LiDAR, ultrasonic sensors, and sensor fusion used in ADAS.
3.	To study major ADAS functions and algorithms like lane warning, cruise control, and collision avoidance.
4.	To learn ADAS architecture, communication methods, RTOS, data processing, and integration issues.
5.	To understand ADAS testing methods, safety standards, validation challenges, and future AI trends.

UNIT–I	**INTRODUCTION TO ADAS**	9
	ADAS Categories: Levels of autonomy in vehicles - ranging from Level 0 (no automation) to Level 5 (full automation) - Core Components of ADAS: Sensors (camera, radar, LiDAR) – actuators - computing platforms – algorithms - System Integration: Integration of various components in ADAS - including communication between sensors, controllers and actuators.	
UNIT–II	**SENSOR TECHNOLOGIES IN ADAS**	9
	Camera Based Systems - Radar and LIDAR Technologies - Ultrasonic Sensors - Sensor Fusion - Challenges and Limitations.	
UNIT– III	**ADAS FUNCTIONS AND ALGORITHM**	9
	Lane Departure Warning - Adaptive Cruise Control - Collision Avoidance Systems - Pedestrian Detection - Traffic Sign Recognition.	
UNIT – IV	**ADAS SYSTEM ARCHITECTURE AND INTEGRATION**	9
	ADAS Architecture - Communication Protocols – GSMA Protocol - Real Time Operating Systems (RTOS) - Data Processing and Machine Learning - Integration Challenges.	
UNIT–V	**TESTING, VALIDATION, AND FUTURE TRENDS IN ADAS**	9
	ADAS Testing Methodologies - Functional Safety and Standards Compliance - Validation Challenges - Machine Learning and AI in ADAS.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the basics of ADAS, levels of vehicle automation, and core system components.	L2 – Understand
CO2	Explain the working of sensor technologies used in ADAS and their	L2 –

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	limitations.	Understand
CO3	Describe key ADAS functions and the algorithms used for driver assistance.	L2 – Understand
CO4	Model and Simulate ADAS system architecture, communication methods, and integration concepts.	L3 – Apply
CO5	Identify testing methods, safety standards, and future trends in ADAS.	L3 – Apply
	REFERENCE BOOKS:	
1.	J. Borenstein, M. Herling, M. Pachter "Advanced Driver Assistance Systems: A Comprehensive Guide", Wiley, 2021.	
2.	D. Makarand, R. Arun "Advanced Vehicle Control Systems: ADAS, Autonomous and Connected Vehicle Technologies", CRC Press, 2021.	
3.	A. Fritzsche, H. Reuss "Advanced Driver Assistance Systems", Springer Vieweg, 2017.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1			3	2	
CO2	1			3	2	
CO3	1			3	2	
CO4	1			3	2	
CO5	1			3	2	
Avg.	**1.0**			**3.0**	**2.0**	
1-Low, 2 -Medium, 3-High.						

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ME23AE601	PROJECT WORK – PHASE I	CP	L	T	P	C
		12	0	0	12	6
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:

1.	To help students identify real-world engineering problems through literature survey and analysis.
2.	To develop the ability to plan and execute an engineering project independently.
3.	To improve skills in technical report writing and project documentation.
4.	To encourage the application of engineering concepts to practical design or case study problems.
5.	To enhance communication and presentation skills through reviews and viva-voce examination.

COURSE CONTENT

	Each student will individually work on a project topic related to engineering design or manufacturing applications. The topic may be theoretical or based on a case study and must be selected after a detailed literature survey. The chosen topic should be approved by the Head of the Department and carried out under the guidance of a faculty member with expertise in the selected area. The student must submit a project proposal for approval and clearly define the problem, objectives, and methodology. During the course, three project review meetings will be conducted by the Project Review Committee to evaluate progress and provide guidance and suggestions for improvement. At the end of the semester, the student must submit a detailed Phase-I report, including the problem definition, literature review, and proposed methodology. The final evaluation will include continuous assessment during reviews and a viva-voce examination conducted by a panel of examiners, including one external examiner.	
	Total(P)	**180 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Identify and define an engineering problem based on a detailed literature survey.	L2 - Understand
CO2	Analyze existing research and technical information related to the selected project topic.	L4 - Analyze
CO3	Design an appropriate methodology to solve the identified engineering problem.	L3 – Apply
CO4	Develop a technical project report with proper structure and documentation.	L6 - Create
CO5	Present and defend the project work effectively during reviews and viva-voce examination.	L3 - Apply

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	3	3	2	3	2
CO2	3	3	3	2	3	2
CO3	3	3	3	2	3	2
CO4	3	3	3	2	3	2
CO5	3	3	3	2	3	2
Avg.	**3.0**	**3.0**	**3.0**	**2.0**	**3.0**	**2.0**
1-Low,2-Medium,3-High						

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ME23AE602	PROJECT WORK – PHASE II	CP	L	T	P	C
		24	0	0	24	12
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:	
1.	To implement and complete the approved project work by applying concepts learned.
2.	To develop the ability to implement, test, and validate engineering solutions.
3.	To enhance research skills through technical writing and paper publication.
4.	To improve problem-solving and decision-making skills through continuous project reviews.
5.	To strengthen technical communication skills through report writing and viva-voce examination.

COURSE CONTENT		
	This phase is a continuation of the Phase I project and will be carried out on the same topic under the guidance of the same supervisor. The student will continue the work based on the approved methodology and complete the project to the satisfaction of the supervisor and the Project Review Committee. Three project review meetings will be conducted during the semester to monitor progress and provide suggestions for improvement. The student is expected to publish at least one research paper related to the project in a national or international conference. At the end of the course, the student must submit a detailed project report to the Head of the Department. The final evaluation will be based on the project report and a viva-voce examination conducted by a panel of examiners, including one external examiner.	
	Total(P)	**360 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Implement the approved project plan and carry out the project work successfully.	L3 - Apply
CO2	Analyze project results and evaluate system performance or outcomes.	L4 - Analyze
CO3	Design and implement improvements based on review committee feedback.	L5 - Evaluate
CO4	Prepare a complete technical report and research paper suitable for publication	L6 - Create
CO5	Present and defend the project work effectively during reviews and viva-voce.	L3 - Apply

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Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	3	3	3	2	3	2
CO2	3	3	3	2	3	2
CO3	3	3	3	2	3	2
CO4	3	3	3	2	3	2
CO5	3	3	3	2	3	2
Avg.	**3.0**	**3.0**	**3.0**	**2.0**	**3.0**	**2.0**
1-Low,2-Medium,3-High						

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PROFESSIONAL ELECTIVES

VERTICAL – I EMBEDDED SYSTEMS AND ECU DESIGN

ME23AE401	RTOS IN MULTI-CORE ENVIRONMENT	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To understand the basics of real-time operating systems and multi-core processors.	
2.	To learn how real-time tasks are scheduled on multi-core systems.	
3.	To practice using synchronization and communication methods for multiple tasks running together.	
4.	To gain hands-on knowledge of RTOS platforms and programming for multi-core systems.	
5.	To study real-time applications and problems related to safety, power, and performance.	
UNIT–I	**INTRODUCTION TO RTOS AND MULTI-CORE ARCHITECTURES**	**9**
	Basics of real-time systems - Hard vs. soft real time constraints - RTOS architecture and components - Multi-core / many-core processors - Overview Multi-core system design challenges – timing – synchronization - scalability - Performance metrics in multi-core real time systems.	
UNIT II	**REAL-TIME SCHEDULING ON MULTI-CORE SYSTEMS**	**9**
	Uniprocessor vs. multiprocessor scheduling - Static and dynamic scheduling techniques - Partitioned vs. global scheduling approaches - Earliest Deadline First (EDF) - Rate Monotonic (RM) scheduling - Load balancing and migration - Timing analysis and schedulable.	
UNIT III	**SYNCHRONIZATION AND COMMUNICATION**	**9**
	Shared memory communication - Task synchronization in real-time environments - Mutual exclusion protocols - Priority inheritance - Priority ceiling – Spinlocks – mutexes - semaphores in multi-core systems - Deadlocks - priority inversion – starvation - Cache coherence and memory models.	
UNIT IV	**RTOS PLATFORMS AND PROGRAMMING ON MULTI-CORE**	**9**
	RTOS architectures for multi-core: asymmetric – symmetric – clustered - Multi-core RTOS examples - FreeRTOS SMP – QNX – VxWorks – RT - Linux - Inter-core communication mechanisms - Task creation – scheduling - synchronization APIs - Performance optimization techniques – Real time debugging and tracing tools.	
UNIT V	**APPLICATIONS, DESIGN ISSUES & CASE STUDIES**	**9**
	Real-time multi-core applications: automotive – avionics – robotics – IoT – telecom – Energy aware scheduling - power management – Safety – reliability - certification standards (ISO 26262, DO-78C) – Real time virtualization and hypervisors - Case studies: AUTOSAR based multi-core - Linux RT PREEMPT - QNX in ADAS.	
	Total(L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the RTOS concepts and challenges in multi-core environments.	L2 - Understand
CO2	Apply scheduling algorithms for real-time tasks on multi-core systems.	L3 - Apply
CO3	Implement synchronization and communication mechanisms in real-time applications.	L3 - Apply
CO4	Build with RTOS platforms supporting multi-core architectures.	L3 - Apply
CO5	Analyze real-time performance, safety, and power issues in practical systems.	L4 - Analyze
	REFERENCE BOOKS:	
1.	Uwe Kiencke, Lars Nielsen, "Automotive Control Systems", Springer, 2nd Edition, 2005.	
2.	Rajib Mall, "Real-Time Systems: Theory and Practice", Pearson Education, 2nd Edition, 2009.	
3.	Jane W. S. Liu, "Real-Time Systems", Prentice Hall, 1st Edition, 2000.	
4.	Klein, Lawrence, "Design of Real-Time Systems", Prentice Hall, 1st Edition, 1996.	
5.	Laplante, Phillip, "Real-Time Systems Design and Analysis", Wiley, 4th Edition, 2011.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1	2	2	1	2
CO2	2	1	3	3	1	3
CO3	2	1	3	3	1	3
CO4	2	1	3	3	1	3
CO5	3	2	3	3	2	3
Avg.	**2.0**	**1.2**	**2.8**	**2.8**	**1.2**	**2.8**
1-Low- 2 -Medium- 3-High.						

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ME23AE402	PARALLEL PROGRAMMING USING MULTI-CORES AND GPUs	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To understand the basic architecture of processors and why multi-core systems are needed.	
2.	To learn how threads work in operating systems and hardware.	
3.	To study the basic concepts of parallel programming and memory systems.	
4.	To understand different synchronization techniques used in parallel programs.	
5.	To learn how to write and manage multi-threaded programs using OpenMP.	
UNIT–I	**MULTI-CORE ARCHITECTURE**	**9**
	Overview of Single core processor - Architecture and its limitations - Architectural Innovations - Need for Multi-core Processor and its Limitations - Classification Multicores - Multicore system software stack.	
UNIT–II	**OVERVIEW OF THREADING**	**9**
	Defining threads – threads inside the OS – threads inside the hardware – Application programming models and threading – virtual environment – Run time virtualization – System virtualization.	
UNIT– III	**FUNDAMENTAL CONCEPTS OF PARALLEL PROGRAMMING**	**9**
	Thread Level Parallelism(TLP) - Instruction Level Parallelism(ILP) – Comparisons - Cache-Hierarchy and Memory level Parallelism - Cache Coherence - Parallel programming models - Shared Memory and Message Passing - Vectorization.	
UNIT – IV	**PARALLEL PROGRAMMING CONSTRUCTS**	**9**
	Synchronization – Critical sections – Deadlock – Semaphores – Locks – Condition variables – Messages – Fence – Barrier – Implementation dependent threading feature.	
UNIT–V	**OPENMP: PORTABLE SOLUTION FOR THREADING**	**9**
	Loop carried dependence – Data-race conditions – Managing shared and private Data – Loop Scheduling and Partitioning – Effective use of reductions – work sharing sections – Using barrier and Nowait – Interleaving single thread and multi-thread execution – Data copy-in and copy-out – Protecting updates of shared variables – OpenMP Library functions – OpenMP environmental variables – multithreading debugging techniques.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the architecture of single-core and multi-core processors.	L2 - Understand

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CO2	Describe how threads work and how virtualization supports them.	L2 - Understand
CO3	Apply basic parallel programming techniques to solve problems.	L3 - Apply
CO4	Use synchronization methods to avoid conflicts in multi-threaded programs.	L3 - Apply
CO5	Write simple parallel programs using OpenMP for better performance.	L3 - Apply
	REFERENCE BOOKS:	
1.	Shameem Akhter, Jason Roberts, "Multi-Core Programming, Increasing Performance through Software Multi-threading", Intel Press, BPB Publications, New Delhi, 2015.	
2.	David B. Kirk, Wen-mei W. Hwu, "Programming Massively Parallel Processors: A hands-on approach", Elesevier, New Delhi, 2015.	
3.	Peter S. Pacheco, "An Introduction to Parallel Programming" Morgan Kaufmann, Elsevier, 2011.	
4.	Sasikumar M., Dinesh Shikhare, Ravi P. Prakash, "Introduction to Parallel Processing", PHI Learning, 2nd Edition, 2014.	
5.	Thomas Rauber, Gudula Runger, "Parallel Programming for Multicore and Cluster Systems", Springer, 3rd Edition, 2023.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	1	1	2	2	1	2
CO2	1	1	2	1	1	2
CO3	2	1	3	2	1	3
CO4	2	1	3	2	1	3
CO5	3	2	3	3	2	3
Avg.	**1.8**	**1.2**	**2.6**	**2.0**	**1.2**	**2.6**
1–Low- 2 –Medium- 3–High.						

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ME23AE403	OPEN SOURCE HARDWARE AND SOFTWARE SYSTEM DESIGN	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To introduce basic programming concepts such as variables, operators, loops, functions, and file handling.	
2.	To learn how to create graphical user interfaces (GUI) and work with web programming methods.	
3.	To understand how to connect and interact with databases for storing and managing data.	
4.	To learn network programming concepts using sockets and implement simple communication programs.	
5.	To gain basic knowledge of Raspberry Pi hardware, setup, and interfacing with sensors and devices.	
UNIT–I	**BASICS**	**9**
	Variable types – basic operators – decision making – loops – strings - Lists – Tuples – Dictionary – Date and Time – Functions – Modules – Files – Exceptions – Classes and Objects.	
UNIT–II	**GUI AND WEB PROGRAMMING**	**9**
	Tkinter Programming – Tkinter Widgets – CGI – Web server support – Environmental variables – GET and POST methods – Passing information using POST method.	
UNIT– III	**DATA BASE ACCESS**	**9**
	MySQLdb – database connection – Creating database table – INSERT – READ – UPDATE – DELETE – COMMIT – ROLEBACK channel - Equalizers - Adaptive beam forming.	
UNIT – IV	**NETWORK PROGRAMMING**	**9**
	Sockets – Server socket – Client Socket – General Socket methods – Sending an HTTP e-mail – Sending an attachment as an email.	
UNIT–V	**RASPBERRY PI FUNDAMENTALS**	**9**
	Architecture – setting up the Raspberry Pi – Interacting with Raspberry command line – Setting up I2C, serial port – Connect Pi to network- Controlling the brightness of LED – Buzzing sound – Switch high power DC source using transistor and relays.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Use basic programming techniques to solve simple problems using functions, files, and classes.	L3 - Apply
CO2	Create simple GUI applications and web programs using standard tools and libraries.	L3 - Apply
CO3	Perform basic database operations like inserting, reading, updating, and deleting data.	L3 - Apply
CO4	Write basic network programs to send and receive data between systems.	L3 - Apply

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CO5	Set up and program a Raspberry Pi to control simple electronic devices and interact with sensors.	L3 - Apply
	REFERENCE BOOKS:	
1.	Richard Blum, Christine Bresnahan, "Python programming for Raspberry Pi in 24 hours", Sams Teach Yourself, Indiana, 2015.	
2.	Simon Monk, "Raspberry Pi Cookbook", O'Reilly Media, 2nd Edition, 2015.	
3.	Mark Lutz, "Learning Python", O'Reilly Media, 5th Edition, 2013.	
4.	Eben Upton, Gareth Halfacree, "Raspberry Pi User Guide", Wiley, 4th Edition, 2016.	
5.	Simon Monk, "Programming the Raspberry Pi: Getting Started with Python", McGraw-Hill Education, 2nd Edition, 2015.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	2	2	1	2
CO2	2	2	2	2	1	2
CO3	2	1	2	2	2	2
CO4	2	1	2	2	1	2
CO5	3	2	3	3	2	3
Avg.	**2.2**	**1.4**	**2.2**	**2.2**	**1.4**	**2.2**

1–Low- 2 –Medium- 3–High.

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ME23AE404	AUTOMOTIVE EMI AND EMC STANDARDS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To introduce the fundamentals of EMC and grounding techniques used in automotive systems.	
2.	To explain basic EMC concepts involving antennas, transmission lines, shielding, and filtering.	
3.	To provide understanding of electromagnetic fields, Maxwell's equations, and EM environments.	
4.	To familiarize students with EMC testing methods, diagnostics, and compliance standards.	
5.	To develop knowledge of EMC design challenges and solutions in automotive applications.	
UNIT–I	**FUNDAMENTALS OF EMC IN AUTOMOTIVE SYSTEMS**	**9**
	Introduction to EMC - System level EMC issues – component and system level - Significance of EMC in automotive electronics - Power and signal return paths – Safety grounding and single-point grounding methods.	
UNIT–II	**BASIC EMC CONCEPTS AND COMPONENTS**	**9**
	Antennas and omni-directional antennas - Transmission lines and shielding - Fourier series and signal spectrum - Capacitors and inductors – ideal vs practical behavior - Filtering techniques - Enclosure shielding and shield discontinuities.	
UNIT– III	**ELECTROMAGNETIC FIELDS AND ENVIRONMENT**	**9**
	Characteristics of electromagnetic environment - Circuit theory vs. electromagnetic field theory - Maxwell's equations - Source regions and field propagation - Polarization of electromagnetic waves.	
UNIT – IV	**EMC MEASUREMENT AND TESTING**	**9**
	MC disciplines and test standards-Radiated emission diagnostics - Conducted emission and immunity tests - Switching transients and their impact - Automotive EMC test methods - compliance requirements.	
UNIT–V	**EMC IN AUTOMOTIVE SYSTEMS AND DESIGN PRACTICES**	**9**
	Cable and harness effects - Filter placement and coupling between wires – Grounding - PCB layout - ferrites and high-frequency emissions -Vehicle - generated radiated emissions - Broadband and narrowband noise - Automotive EMC case studies: ABS – ignition - fuel systems - Shielding techniques and immunity design - Industry practices and design guidelines.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain system-level EMC issues and grounding strategies in vehicles.	L2 – Understand
CO2	Apply EMC principles involving antennas, shielding, and filtering components.	L3 – Apply

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CO3	Apply electromagnetic fields using theoretical models and Maxwell's equations.	L3 – Apply
CO4	Evaluate EMC tests and diagnostics for radiated and conducted emissions.	L3 – Apply
CO5	Implement EMI reduction techniques, grounding, PCB layout, and shielding in automotive systems.	L3 – Apply
	REFERENCE BOOKS:	
1.	Terence Rybak, Mark steffka, "Automotive Electromagnetic compatibility", Kluver Academic Publishers, 2015.	
2.	Balcells, D. González, J. Gago, "EMC Design in Industrial Systems", Marcombo / Wiley, 2015.	
3.	D.A.Weston, "Electromagnetic Compatibility: Principles and Applications", Marcel Dekker, 2nd Revised and Expanded Edition, 2001.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	2	2
CO2	2	1	3	3	2	2
CO3	3	2	3	2	1	1
CO4	2	2	3	3	2	3
CO5	3	2	3	3	3	3
Avg.	**2.4**	**1.6**	**3.0**	**2.6**	**2.0**	**2.2**
1-Low- 2 –Medium- 3–High.						

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ME23AE405	AUTOMOTIVE POWER ELECTRONICS AND MOTOR DRIVES	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To learn the basics of power electronic devices and their protection.	
2.	To understand how different converters work with various loads.	
3.	To learn the working of inverters, choppers and PWM methods.	
4.	To understand ultracapacitors, their sizing and use with batteries.	
5.	To learn how different motors are controlled in automotive drive systems.	
UNIT–I	**POWER ELECTRONICS FUNDAMENTALS**	**9**
	Introduction to power electronics - Structure- operation and characteristics of automotive semiconductor devices – SCR - Power Transistor - Power MOSFET – IGBT - Turn-on and turn-off circuits - Series and parallel operation of SCR - Protection circuits - Snubber circuit design.	
UNIT–II	**POWER CONVERTERS**	**9**
	Half-wave controlled converters with R, RL, RLE loads - Fully controlled converters with R, RL, RLE loads - Three-phase half-wave controlled converters with R, RL loads - Three-phase fully controlled converters with R, RL loads.	
UNIT– III	**INVERTERS AND CHOPPERS**	**9**
	Voltage Source Inverters (120° and 180° conduction modes) - Current Source Inverters -PWM techniques - Step-up and step-down choppers - Types of choppers and their applications.	
UNIT – IV	**ENERGY STORAGE SYSTEMS (ULTRACAPACITORS)**	**9**
	Theory of electronic double - layer capacitance - Ultracapacitor modelling and cell balancing - Sizing criteria for ultracapacitors - Converter interfaces - Hybrid systems: ultracapacitors combined with batteries.	
UNIT–V	**AUTOMOTIVE MOTOR DRIVES**	**9**
	Methods of speed control - Induction motor control - DC motor control - BLDC motor: construction – characteristics – operation - Open-loop and closed-loop control using speed and current sensors - Switched Reluctance Motor (SRM): construction – operation - applications.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	**BLOOM'S Taxonomy**
CO1	Use the characteristics of power electronic devices to implement basic switching and protection circuits.	L3 - Apply
CO2	Demonstrate the operation of single-phase and three-phase converters for different load conditions.	L3 - Apply
CO3	Implement inverter, chopper, and PWM techniques for DC–DC and DC–AC power conversion.	L3 - Apply
CO4	Calculate ultracapacitor sizing and use models to integrate them with battery systems.	L3 - Apply

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CO5	Apply suitable speed control techniques to operate DC, induction, BLDC, and SRM motors.	L3 - Apply
	REFERENCE BOOKS:	
1.	P.S. Bimbhra, "Power Electronics", Khanna Publishers, 14th edition, 2015.	
2.	Ali Emadi, "Handbook of Automotive power electronics and motor Drives", CRC Press, 2015.	
3.	Bimal K. Bose, "Power Electronics and Motor Drives: Advances and Trends", Elsevier, 1st Edition, 2006.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	3	1	2
CO2	2	1	3	3	1	2
CO3	3	2	3	3	2	2
CO4	3	2	3	2	3	2
CO5	3	2	3	3	2	2
Avg.	**2.6**	**1.6**	**3.0**	**2.8**	**1.8**	**2.0**
1-Low- 2 -Medium- 3-High.						

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ME23AE406	ALTERNATIVE DRIVES, TRACTION AND CONTROLS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:

1.	To understand how automotive electrical and electronic systems work in modern vehicles.
2.	To learn about alternative powertrains and hybrid vehicle systems, and their role in reducing pollution and saving energy.
3.	To study and compare vehicle performance, especially differences between IC engines and electric motors.
4.	To know different energy storage and conversion systems, like batteries, fuel cells, and charging methods for EVs.
5.	To use electric motor control techniques and energy recovery methods to improve vehicle performance and efficiency.

UNIT–I	**AUTOMOTIVE ELECTRICAL & ELECTRONIC SYSTEMS**	9
	Electrical systems and circuits-Starting systems - Ignition systems - Lighting - accessories and safety systems - Electromagnetic interference (EMI) and compatibility (EMC) fundamentals.	
UNIT–II	**ALTERNATE POWERTRAINS & HYBRID VEHICLE TECHNOLOGY**	9
	Need for alternate propulsion technologies - Emissions from IC engine – based transportation and regulatory standards - Availability projections for non-renewable energy resources - Alternate technologies for reducing urban pollution and extending resources - Importance and basic principles of Hybrid Electric Vehicles - Architectures of HEVs: series, parallel, and series–parallel - Overview of drive train ratings and component sizing.	
UNIT– III	**VEHICLE PROPULSION SYSTEMS**	9
	Components determining traction torque - Vehicle performance parameters: speed, acceleration - Fuel economy and performance in IC engine vehicles – Torque – speed characteristics of IC engines - Electric vs. IC engines for vehicle propulsion - Basics of electric vehicles - Types of electric motors – Speed – torque characteristics of EV motors.	
UNIT – IV	**ENERGY STORAGE AND ENERGY CONVERSION TECHNOLOGIES**	9
	Batteries for EVs: Lead acid – NiMH – Lithium ion - Comparison of battery technologies - Battery Management Systems / Energy Management Systems - Wireless and fast charging systems - Super capacitors - Fuel cell fundamentals - Solar energy conversion for EVs.	
UNIT–V	**ELECTRIC MOTORS, DRIVES & CONTROL TECHNIQUES**	9
	C motors – principle and control methods - Induction motor drives – speed control methods - Constant V/f control - Vector control method – inverter for vector control - BLDC motors – basic principles - performance and control - Sensorless techniques for BLDC motor drives - Regenerative braking with electric drives - Four-quadrant operation - Optimizing energy recovery in electric drive systems.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply knowledge to interpret automotive electrical systems, including ignition, starting, lighting, and EMI/EMC aspects.	L3 - Apply
CO2	Use concepts of alternate propulsion to compare hybrid vehicle designs based on energy, environment, and layout.	L3 - Apply
CO3	Analyze vehicle performance by relating torque, speed, acceleration, and fuel economy for different power sources.	L4 - Analyze
CO4	Compare battery types and illustrate how energy storage and management work in electric and hybrid vehicles.	L4 - Analyze
CO5	Evaluate motor drives, control techniques, and regenerative braking to improve performance and energy efficiency.	L3 - Apply
	REFERENCE BOOKS:	
1.	Mehrdad Ehsani, Yimin Gao, Sebastien Gay and Ali Emadi, "Modern Electric, Hybrid Electric and Fuel Cell Vehicles", CRC Press, 2015.	
2.	Iqbal Husain, "Electric & Hybrid Vehicles", CRC Press, 2015.	
3.	Ronald K Jurgen, "Automotive Electronics Handbook", McGraw-Hill, 1999.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2	1	3	3	1	2
CO2	2	1	3	2	3	2
CO3	3	2	3	3	2	2
CO4	2	1	3	2	3	2
CO5	3	2	3	3	3	2
Avg.	**2.4**	**1.4**	**3.0**	**2.6**	**2.4**	**2.0**
1-Low- 2 -Medium- 3-High.						

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ME23AE407	AUTOSAR AND ISO STANDARDS FOR AUTOMOTIVE SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To understand AUTOSAR basic software standards and error-handling concepts.	
2.	To learn AUTOSAR communication stack, TTCAN, and network management.	
3.	To understand AUTOSAR implementation, memory mapping, and system services.	
4.	To learn ISO/TS 16949 communication principles, noise reduction, and grounding methods.	
5.	To understand ISO 26262 functional safety concepts, ASIL, and implementation practices.	
UNIT–I	**AUTOSAR BASICS & SOFTWARE STANDARDS**	**9**
	General requirements of AUTOSAR Basic Software (BSW) modules - Functional operation requirements - Fault operation and error detection mechanisms.	
UNIT–II	**AUTOSAR COMMUNICATION STACK**	**9**
	Network Management - TTCAN (Time-Triggered CAN) Interface standards - TTCAN Drivers - AUTOSAR communication principles.	
UNIT– III	**AUTOSAR IMPLEMENTATION & SYSTEM SERVICES**	**9**
	AUTOSAR Platform Types - Memory Mapping - Watchdog Manager - Synchronized Time Base Manager.	
UNIT – IV	**ISO/TS 16949 & AUTOMOTIVE COMMUNICATION BASICS**	**9**
	Data transmission systems - Pulse code formats - Modulation techniques -Telemetry concepts - Noise sources and interference - Noise reduction techniques - Signal grounding - shield grounding – Capacitive - magnetic-optical isolation.	
UNIT–V	**ISO 26262 FUNCTIONAL SAFETY**	**9**
	Structure and parts of ISO 26262 - Vocabulary and functional safety management - Concept phase requirements - Product development (System, Hardware, Software levels) - Production, operation, and supporting processes – ASIL oriented and safety analysis - Hazard analysis & risk assessment - Safety goals, preliminary architecture - Functional safety concept – Illustrative case studies.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	**BLOOM'S Taxonomy**
CO1	Apply AUTOSAR basic software requirements to identify functions, faults, and error-detection needs.	L3 - Apply
CO2	Use AUTOSAR communication standards to configure network management and TTCAN drivers.	L3 - Apply

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CO3	Implement AUTOSAR platform concepts, memory mapping, and key system services.	L3 - Apply
CO4	Apply ISO/TS 16949 principles to analyze data transmission, grounding, and noise-reduction techniques.	L3 - Apply
CO5	Use ISO 26262 guidelines to perform hazard analysis, assess risks, and develop functional safety concepts.	L3 - Apply
	REFERENCE BOOKS:	
1.	Miroslaw Staron, "Automotive Software Architectures", Springer International Publishing, 2nd Edition, 2021.	
2.	David Hoyle, "Automotive Quality Systems", Butterworth-Heinemann, 4th Edition, 2015.	
3.	Dominique Schreurs, Eli Torkos, and Bart Nauwelaers, "Automotive EMC Standards and Testing", Wiley, 1st Edition, 2017.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	3	1	2
CO2	2	1	3	3	1	2
CO3	3	2	3	3	1	2
CO4	2	1	2	3	2	1
CO5	3	2	3	3	3	2
Avg.	**2.4**	**1.4**	**2.8**	**3.0**	**1.6**	**1.8**
1–Low- 2 –Medium- 3–High.						

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ME23AE408	VEHICLE ELECTRIFICATION AND POWERTRAIN INTEGRATION	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:

1.	To understand the fundamentals of vehicle electrification, powertrain architecture, and key components.
2.	To learn the working principles and control of electric machines and power electronic converters.
3.	To understand battery technologies, BMS functions, and thermal management in energy storage systems.
4.	To study hybrid/electric powertrain configurations and supervisory control strategies.
5.	To understand EV charging systems, standards, and emerging technologies in electrified mobility.

UNIT–I	**BASICS OF VEHICLE ELECTRIFICATION**	**9**
	Types of electrified vehicles: HEV, PHEV, BEV - EV system architecture and main components - Basics of electric motors and their characteristics - Energy flow in electric powertrains - Standards and safety basics for EVs.	
UNIT–II	**ELECTRIC POWERTRAIN ARCHITECTURE**	**9**
	Powertrain layouts: series – parallel – series and parallel - Electric drive units: motor – inverter – transmission - Motor control fundamentals (FOC,PWM) - Thermal management and packaging - Drive cycle analysis and performance basics.	
UNIT–III	**BATTERIES AND ENERGY STORAGE**	**9**
	Battery types and characteristics (Li-ion variants) - Battery pack configuration and thermal management - SOC, SOH basics - Battery Management System (BMS) essentials - Charging methods and standards.	
UNIT–IV	**POWER ELECTRONICS FOR EVS**	**9**
	Introduction to power converters (DC–DC inverters) - Power semiconductor devices (MOSFET, IGBT, SiC) - On-board charger basics - Regenerative braking principles – High voltage system safety.	
UNIT– V	**VEHICLE CONTROL, COMMUNICATION & DIAGNOSTICS**	**9**
	Vehicle Control Unit (VCU) basics – CAN – LIN - Ethernet communication - Functional safety (ISO 26262) overview - Basic diagnostics and fault detection - Trends: wireless charging - V2G – software defined vehicles.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course the students will be able to:	BLOOM'S Taxonomy
CO1	Describe EV powertrain architecture and major electrification components.	L2 - Understand
CO2	Explain different electric machines and power converters for EV applications.	L2 - Understand
CO3	Evaluate battery performance, safety, and BMS operations.	L3 - Apply
CO4	Model the powertrain integration and control strategies used in EVs/HEVs.	L3 - Apply
CO5	Organize EV charging methods, standards, and emerging technologies.	L3 - Apply
	REFERENCE BOOKS:	
1.	Iqbal Husain, "Electric and Hybrid Vehicles: Design Fundamentals", CRC Press, 2nd Edition, 2010.	
2.	Mehrdad Ehsani, Yimin Gao, Sebastien E. Gay, "Modern Electric, Hybrid Electric, and Fuel Cell Vehicles", CRC Press, 2nd Edition, 2009.	
3.	James Larminie, John Lowry, "Electric Vehicle Technology Explained", Wiley, 2nd Edition, 2012.	
4.	Ali Emadi, Sheldon Williamson, Alireza Khaligh, "Advanced Electric Drive Vehicles", CRC Press, 1st Edition, 2014.	
5.	V. Ganesan, "Automotive Powertrains and Electric Vehicles", McGraw Hill, 1st Edition, 2019.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	2	1
CO2	2	1	3	3	2	2
CO3	2	1	3	3	3	1
CO4	3	2	3	3	2	2
CO5	2	1	2	2	3	2
Avg.	**2.2**	**1.2**	**2.8**	**2.6**	**2.4**	**1.6**

1-Low- 2 -Medium- 3-High.

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VERTICAL – II AUTOMOTIVE COMMUNICATION, NETWORKING AND DIAGNOSTICS

ME23AE409	**VEHICULAR INFORMATION AND COMMUNICATION SYSTEMS**	**CP**	**L**	**T**	**P**	**C**
		3	**3**	**0**	**0**	**3**
Programme & Branch	**M.E. AUTOMOTIVE ELECTRONICS**	**Version: 1.0**				

Course Objectives:		
1.	To understand how ECUs work and control different vehicle functions.	
2.	To learn how vehicle communication networks like CAN, LIN, and FlexRay operate.	
3.	To know how driver information systems and displays help drivers.	
4.	To understand how vehicles record data, assist parking, and track position.	
5.	To learn about in-car entertainment systems and how route planning is done.	
UNIT–I	**ECU BASICS & VEHICLE DATA PROCESSING**	**9**
	Data processing requirements in motor vehicles - Electronic Control Unit (ECU): Architecture & Functions - Vehicle control architecture (CARTRONIC overview).	
UNIT–II	**AUTOMOTIVE NETWORKING & COMMUNICATION SYSTEMS**	**9**
	Cross-system functions and communication needs - Requirements and classification of in-vehicle bus systems - Applications of bus systems (CAN, LIN, FlexRay examples) - Network coupling and gateway concepts.	
UNIT– III	**DRIVER INFORMATION & INSTRUMENTATION SYSTEMS**	**9**
	Information and communication domains in vehicles - Driver information systems - Instrument clusters - Display technologies and modern HMI elements.	
UNIT – IV	**ECU RECORDING, PARKING & POSITIONING SYSTEMS**	**9**
	Legal requirements for ECU data recording - Event data recorders: design variations - Parking assistance systems: ultrasonic sensor – based aids - Positioning: Dead reckoning - GPS sensor fusion - Map matching - conventional - fuzzy logic – map - aided calibration.	
UNIT–V	**ROUTE PLANNING, SOUND & INFOTAINMENT SYSTEMS**	**9**
	Radio & audio systems: tuner types - digital receivers – antennas - Reception quality and enhancement techniques - Route planning algorithms: shortest path – heuristic – bidirectional - hierarchical search - Route guidance: on route – off route - dynamic guidance.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain vehicle data processing concepts in relation to ECU architecture, functions, and control structure.	L2 - Understand
CO2	Identify and classify in-vehicle communication systems such as CAN, LIN, and FlexRay.	L3 - Apply
CO3	Make use of driver information systems to understand instrument clusters, displays, and HMI elements.	L3 - Apply

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CO4	Utilize ECU data recording concepts for parking assistance and vehicle positioning using GPS and sensor fusion.	L3 - Apply
CO5	Apply route planning algorithms and infotainment systems for effective vehicle guidance and user experience.	L3 - Apply
	REFERENCE BOOKS:	
1.	Bosch, "Automotive Handbook", 8th Edition, SAE publication, 2015.	
2.	Ljubo Vlacic, Michel Parent, Fumio Harashima, "Intelligent Vehicle Technologies: Theory and Applications", Butterworth-Heinemann, 2001.	
3.	Yilin Zhao, "Vehicle location and Navigation Systems", Artech House, 2016.	
4.	Joseph, "Perspectives on Intelligent Transportation Systems (ITS)", NewYork, Springer, 2010.	
5.	Mashrur A. Chowdhury, Adel Sadek, "Fundamentals of Intelligent Transportation Systems Planning", Artech House, 2003.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	1	1
CO2	2	1	3	3	1	2
CO3	1	1	2	2	1	1
CO4	2	1	3	2	2	2
CO5	1	1	2	2	1	1
Avg.	**1.6**	**1.0**	**2.6**	**2.2**	**1.2**	**1.4**
1-Low- 2 -Medium- 3-High.						

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ME23AE410	AUTOMOTIVE FAULT DIAGNOSTICS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To introduce the principles and processes of automotive diagnostics.	
2.	To develop understanding of diagnostic tools, instruments, and data interpretation.	
3.	To enable learners to use oscilloscopes for analyzing automotive components and signals.	
4.	To provide knowledge on on-board diagnostics and its application in fault detection.	
5.	To train students in diagnosing faults in engine, chassis, electrical, and safety systems.	
UNIT–I	FUNDAMENTALS OF AUTOMOTIVE DIAGNOSTICS	9
	Diagnostic techniques and diagnostic process – Paper based diagnostics - data sources - Mechanical and electrical diagnostic methods - Fault codes - on-board and off-board diagnostics.	
UNIT–II	DIAGNOSTIC TOOLS AND EQUIPMENT	9
	Basic diagnostic tools and workshop instrumentation – Oscilloscopes - scanners - fault-code readers - Engine analyzers and test procedures - Interpretation of diagnostic data.	
UNIT– III	OSCILLOSCOPE-BASED DIAGNOSTICS	9
	Oscilloscopes in automotive diagnostics - Sensor signal analysis - Actuator performance evaluation - Ignition system diagnostics and waveform interpretation.	
UNIT – IV	ON-BOARD DIAGNOSTICS (OBD) SYSTEMS	9
	Principles of OBD systems - Petrol/Gasoline OBD monitoring - OBD data interpretation and fault analysis - OBD regulations and standards.	
UNIT–V	AUTOMOTIVE SYSTEM DIAGNOSTICS	9
	Engine system diagnostics: fuel - ignition – emission - injection management - Air intake - exhaust – cooling - lubrication systems - Battery - starting - charging system faults – Brake - ABS - traction control - steering and suspension diagnostics - Electrical system diagnostics: multiplexing - lighting - auxiliary systems - Instrumentation - HVAC - infotainment - cruise control - airbag and safety systems.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain diagnostic techniques, fault codes, and diagnostic processes used in vehicles.	L2 - Understand
CO2	Use diagnostic tools and analyze data for fault identification.	L4 - Analyze
CO3	Apply oscilloscope techniques to evaluate sensors, actuators, and ignition systems.	L3 - Apply
CO4	Interpret OBD data and perform fault analysis on automotive systems.	L3 - Apply
CO5	Diagnose faults in engine, chassis, electrical, and safety systems using	L3 - Apply

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	systematic methods.	
	REFERENCE BOOKS:	
1.	Tom Denton, "Automotive Technician Training", Taylor and Francis, New York, 2015.	
2.	Tom Denton, "Automobile Electrical and Electronic Systems: Automotive Technology - Vehicle Maintenance and Repair", Fourth Edition, Elsevier, New York, 2015.	
3.	Robert Bosch GmbH, "Automotive Handbook", 10th Edition, Wiley, 2018.	
4.	Tom Denton, "Advanced Automotive Fault Diagnosis", 4th Edition, Routledge, 2014.	
5.	James D. Halderman, "Automotive Technology: Principles, Diagnosis and Service", 5th Edition, Pearson, 2016.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	1	1
CO2	3	2	3	3	1	2
CO3	2	1	3	3	1	2
CO4	2	2	3	3	2	2
CO5	3	2	3	3	2	3
Avg.	**2.4**	**1.6**	**3.0**	**2.8**	**1.4**	**2.0**

1-Low- 2 -Medium- 3-High.

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ME23AE411	AUTOMOTIVE TELEMATICS AND IoT CONNECTIVITY	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:

1.	To understand the fundamentals of telematics and connected vehicle architectures.
2.	To study vehicle data acquisition, processing, and telematics services.
3.	To explore IoT protocols, platforms, and their automotive applications.
4.	To learn V2X communication technologies and their applications in connected and autonomous vehicles.
5.	To analyze cybersecurity, data privacy, and future trends in automotive telematics and IoT.

UNIT–I	**INTRODUCTION TO TELEMATICS & VEHICLE CONNECTIVITY**	**9**
	Evolution of telematics and connected vehicles - Architecture of connected vehicle systems - Core components: sensors – ECUs - HMI - cloud – apps - Vehicle communication requirements: latency – bandwidth - reliability - security - Telematics Control Unit (TCU): hardware and software - GPS/GNSS fundamentals and vehicle positioning - Cellular standards: 2G - 3G - 4G LTE - 5G NR - Emerging connectivity: NB-IoT - LTE-M - DSRC.	
UNIT–II	**VEHICLE DATA ACQUISITION, PROCESSING AND TELEMATICS SERVICES**	**9**
	Vehicle data sources: CAN - LIN – FlexRay - Ethernet -Cloud data transmission models: push- pull – streaming - Data logging and storage architecture - Vehicle telematics services: Navigation and routing - Remote vehicle diagnostics - Fleet management systems - Predictive maintenance - Driver monitoring and scoring - Over-the-air updates (OTA) - Data compression - edge computing for vehicles.	
UNIT– III	**INTERNET OF THINGS (IoT) IN AUTOMOTIVE APPLICATIONS**	**9**
	IoT fundamentals: architecture - protocols – platforms – MQTT – CoAP - HTTP - WebSockets - Cloud frameworks: AWS IoT - Azure IoT - Google IoT - IoT gateways and edge devices - IoT analytics and dashboards - Vehicle-to-Cloud (V2C) communication - Digital twin concepts in automotive - Case study: Remote health monitoring.	
UNIT – IV	**VEHICLE-TO-EVERYTHING (V2X) COMMUNICATION**	**9**
	V2X fundamentals and architecture - DSRC vs C-V2X (LTE-V, 5G-V2X) - V2V, V2I, V2P, V2N communication models - Message sets: BSM, CAM, DENM – Multi access edge computing (MEC) - Telematics for ADAS and autonomous driving - V2X applications: intersection safety - collision avoidance - cooperative driving - Testing and validation of V2X systems.	
UNIT–V	**CYBERSECURITY AND DATA PRIVACY**	**9**
	Automotive cybersecurity threats and attack surfaces - Secure communication protocols: TLS - IPSec - Authentication and encryption in IoT/vehicles - Secure OTA software updates - Privacy and data compliance: GDPR, ISO 21434 - Blockchain for automotive data security - AI and ML for telematics analytics - Software defined vehicles.	

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	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Describe the architecture and components of telematics and connected vehicles.	L2 - Understand
CO2	Explain vehicle data acquisition, storage, and telematics services.	L2 - Understand
CO3	Demonstrate basic IoT communication protocols and cloud integration for vehicles.	L3 - Apply
CO4	Compare V2X communication methods and identify their applications in ADAS and autonomous driving.	L3 - Apply
CO5	Apply cybersecurity and data privacy principles to secure telematics and IoT systems.	L3 - Apply
	REFERENCE BOOKS:	
1.	William Buchanan, "Connected Vehicle Systems: Automotive Telematics", Wiley, 1st Edition, 2020.	
2.	Raj Kamal, "Internet of Things: Architecture and Design Principles", McGraw Hill, 2nd Edition, 2017.	
3.	Alessandro Bazzi, "V2X Communication for Vehicular Networks", Springer, 1st Edition, 2019.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	1	1
CO2	2	1	3	2	2	1
CO3	3	2	3	3	2	2
CO4	3	2	3	3	2	2
CO5	3	2	3	3	3	2
Avg.	**2.6**	**1.6**	**3.0**	**2.6**	**2.0**	**1.6**

1-Low- 2 -Medium- 3-High.

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ME23AE412	**DATA ACQUISITION AND SIGNAL CONDITIONING**	**CP**	**L**	**T**	**P**	**C**
		3	**3**	**0**	**0**	**3**
Programme & Branch	**M.E. AUTOMOTIVE ELECTRONICS**	**Version: 1.0**				

Course Objectives:		
1.	To understand the basics of amplifiers and linear integrated circuits.	
2.	To learn how advanced amplifiers and nonlinear circuits work in real applications.	
3.	To understand how signals are sampled, converted, and processed using ADCs and DACs.	
4.	To learn how data is transmitted and how to reduce noise and interference in circuits.	
5.	To understand data acquisition systems and common interfacing/communication standards.	
UNIT–I	**BASIC LINEAR INTEGRATED CIRCUITS & AMPLIFIERS**	**9**
	Introduction to amplifiers and amplifier parameters - Operational amplifiers and characteristics - Differential amplifiers - Instrumentation amplifiers - Basics of carrier amplifiers - lock-in amplifiers and chopper amplifiers.	
UNIT–II	**ADVANCED AMPLIFIERS & NONLINEAR PROCESSING**	**9**
	Low drift amplifiers - electrometer and transimpedance amplifiers -Charge amplifiers and isolation amplifiers - Non-linear signal conditioning: limiting – clipping - logarithmic - exponential amplifiers - Analog linearization techniques - Noise in amplifiers - noise and drift in resistors.	
UNIT– III	**SIGNAL CONVERSION TECHNIQUES**	**9**
	Voltage-to-frequency and capacitance-to-period converters - Frequency-to-code conversion - Sampling concepts and pre-filtering - Sample and Hold circuits - ADCs and DACs – types and architectures - Multiplexers and demultiplexers.	
UNIT – IV	**DATA TRANSMISSION & INTERFERENCE CONTROL**	**9**
	Data transmission systems - Pulse code formats and modulation techniques - Telemetry and remote sensing basics - Noise sources interference mechanisms and reduction methods - Grounding strategies: signal grounding, shield grounding - Capacitive, magnetic, and optical isolation methods.	
UNIT–V	**DATA ACQUISITION & INTERFACING STANDARDS**	**9**
	Data Acquisition Systems (DAS) – structure and operation - DAS boards and interfacing issues - Software drivers and data loggers - Time-division multiplexed data acquisition – multi channel errors & error protection - Instrument communication standards: GPIB (IEEE-488) - RS-232C - 3USB-4–20 mA current loop - Serial communication protocols.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the functioning and design principles of basic and advanced amplifiers used in automotive and sensor applications.	L2 - Understand
CO2	Apply non-linear signal processing techniques and noise reduction methods to improve signal conditioning.	L3 - Apply
CO3	Design and analyse signal conversion circuits such as ADC, DAC, V-F converters, sampling systems, and multiplexers.	L3 - Apply
CO4	Evaluate data transmission methods, interference effects, and implement proper grounding and shielding strategies.	L3 - Apply
CO5	Develop and interface data acquisition systems using DAS boards, software tools, and communication standards such as GPIB, RS-232C, USB, and current loop systems.	L3 - Apply
	REFERENCE BOOKS:	
1.	Pallas Areny. R, Webster. J. G, "Sensors and Signal conditioning", John Wiley and Sons, 2nd Edition, 2015.	
2.	Jacob Fraden, "Handbook of Modern Sensors: physics, Designs and Applications", Springer, 2nd Edition, 2015	
3.	Taylor, H. Rosemary, "Data Acquisition for Sensor Systems", Kluwer Academic Publishers Group, 1997.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	3	1	1
CO2	2	1	3	3	1	1
CO3	3	2	3	3	1	2
CO4	2	1	2	3	2	2
CO5	3	2	3	3	2	3
Avg.	**2.4**	**1.4**	**2.8**	**3.0**	**1.4**	**1.8**

1-Low- 2 -Medium- 3-High.

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ME23AE413	MACHINE VISION SYSTEM FOR AUTOMOTIVE	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:

1.	To provide knowledge on the fundamental components and architecture of computer vision systems.
2.	To develop skills in applying digital image processing techniques for image enhancement and analysis.
3.	To enable students to perform segmentation and feature extraction on digital images.
4.	To train students to analyze, classify, and recognize objects using statistical and computational methods.
5.	To encourage the application of computer vision techniques in industrial inspection and automation systems.

UNIT–I	**FUNDAMENTALS OF COMPUTER VISION SYSTEMS**	**9**
	Artificial intelligence for vision - Image processing and machine vision basics - System architecture and components - Illumination and sensors - Basic optics - camera sensors - camera interfaces - Sampling - quantization - inter-pixel distance - adjacency-Image acquisition hardware and speed considerations.	
UNIT–II	**DIGITAL IMAGE PROCESSING TECHNIQUES**	**9**
	Point operations - contrast stretching - thresholding-Noise reduction - background subtraction - Neighborhood operations and convolution - Geometric operations: warping- interpolation – registration - Morphology: erosion - dilation- opening- closing-Structuring elements - gray-scale morphology.	
UNIT– III	**IMAGE SEGMENTATION AND FEATURE EXTRACTION**	**9**
	Region and boundary based segmentation - Thresholding techniques: global, local, dynamic - Edge detection: gradient and difference based methods - Region growing – quadtree - Boundary detection - contour following – Graph based segmentation - dynamic programming.	
UNIT – IV	**IMAGE ANALYSIS AND CLASSIFICATION**	**9**
	Inspection - localization - identification - Template matching - simple feature extraction - Classification using Bayes' rule - Hough transform - generalized Hough transform - Histogram-based analysis.	
UNIT–V	**SHAPE DESCRIPTION AND INDUSTRIAL VISION APPLICATIONS**	**9**
	Shape descriptor taxonomy - Boundary-based features - chain code - Region based features - internal descriptors - Industrial applications: Object detection and defect detection - Assembly verification - Component classification.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain the architecture of computer vision systems, [illegible]	

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	acquisition techniques.	Understand
CO2	Apply digital image processing operations to enhance and transform images.	L3 - Apply
CO3	Perform segmentation and extract features for object representation.	L3 - Apply
CO4	Analyze and classify image patterns using computational methods.	L4 - Analyze
CO5	Use shape descriptors and vision algorithms to solve industrial inspection problems.	L3 - Apply
	REFERENCE BOOKS:	
1.	E. R. Davies, "Computer and Machine Vision: Theory, Algorithms, Practicalities", 4^{th} Edition, Academic Press, 2012.	
2.	Bruce G. Batchelor, Paul F. Whelan, "Intelligent Vision systems for Industry", Springer, London, 2015.	
3.	Rafael C. Gonzalez, Richard E. Woods, "Digital Image Processing", 4^{th} Edition, Pearson Education, 2018.	
4.	Sonka, Hlavac, Boyle, "Image Processing, Analysis, and Machine Vision", Cengage Learning, 2015.	
5.	Linda G. Shapiro, George C. Stockman, "Computer Vision", Prentice Hall, 2002.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	1	1
CO2	2	1	3	3	1	1
CO3	2	1	3	3	1	1
CO4	3	2	3	3	2	2
CO5	3	2	3	3	2	2
Avg.	**2.4**	**1.4**	**3.0**	**2.8**	**1.4**	**1.4**

1-Low- 2 -Medium- 3-High.

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ME23AE414	EMISSION CONTROL AND DIAGNOSIS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To explain the sources, formation, and effects of automotive emissions.	
2.	To provide knowledge of emission formation and control strategies in SI engines.	
3.	To understand emission characteristics and reduction methods in CI engines.	
4.	To develop skills in exhaust emissions measurement and interpretation.	
5.	To familiarize students with global emission standards and regulatory requirements.	
UNIT–I	AUTOMOTIVE EMISSIONS AND ENVIRONMENTAL IMPACT	9
	Sources of air pollution - Emissions from automobiles: formation and types - Effects of pollutants on environment and human health - Emission control techniques: fuel modification – after treatment devices - Automotive waste management and recycling (vehicles, tyres) - Emission standards overview.	
UNIT–II	EMISSION FORMATION IN SI ENGINES AND CONTROL TECHNIQUES	9
	Emission formation in SI engines: CO, CO_2, HC, NOx, smoke - Influence of design and operating variables - Emission control technologies: Catalytic converters - Charcoal canister - Positive crankcase ventilation - Secondary air injection - Thermal reactor - Laser assisted combustion - CCS (computer-controlled systems).	
UNIT– III	EMISSION FORMATION IN CI ENGINES AND CONTROL TECHNOLOGIES	9
	Emission formation in CI engines: smoke types – Nox – soot - sulphur particulates - Combustion delay: physical and chemical - Influence of operating variables - Low-emission technologies: Fumigation - split injection - Catalytic coating - EGR – HCCI - SCR - particulate traps - Fuel additives - cetane number effects.	
UNIT – IV	EXHAUST GASES, MEASUREMENT AND TEST CYCLES	9
	Combustion products and exhaust gas composition - Properties of exhaust gas components - Exhaust gas analysis and measurement devices - Diesel smoke emission test - evaporative emission test - Test cycles for passenger cars and commercial vehicles: US – EU - Japan standards.	
UNIT–V	EMISSION LEGISLATION AND REGULATORY FRAMEWORK	9
	Global emission control legislation: CARB – EPA – EU - Japan - Regulatory requirements for petrol and diesel engines - Legislative trends and compliance challenges.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: Upon completion of this course the students will be able to:	**BLOOM'S Taxonomy**
CO1	Describe automotive emissions, their effects, and basic control strategies.	L2 - Understand
CO2	Explain emission formation in SI engines and evaluate suitable emission control techniques.	L2 - Understand

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CO3	Analyze emission formation in CI engines and apply appropriate control methods.	L4 - Analyze
CO4	Perform exhaust emission measurements and interpret test results.	L3 - Apply
CO5	Interpret global emission legislation and relate standards to vehicle compliance.	L3 - Apply
	REFERENCE BOOKS:	
1.	G.P.Springer, D.J.Patterson, "Engine Emissions, Pollutant formation", Plenum Press, New York, 1986.	
2.	D.J.Patterson, N.A.Henin, "Emission from Combustion Engine and their control", AnnaArbor Science Publication, 1985.	
3.	John B. Heywood, "Internal Combustion Engine Fundamentals", McGraw Hill, 2nd Edition, 2018.	
4.	Ganesan V, "Internal Combustion Engines", McGraw Hill, 4th Edition, 2017.	
5.	Crouse, Anglin, "Automotive Emission Control", McGraw Hill company, New York, 1993.	

Mapping of COs with POs						
COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2	1	3	2	3	1
CO2	2	1	3	2	2	1
CO3	3	1	3	3	2	2
CO4	3	2	2	3	2	2
CO5	2	2	2	1	3	2
Avg.	**2.4**	**1.4**	**2.6**	**2.2**	**2.4**	**1.6**
1-Low- 2 -Medium- 3-High.						

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ME23AE415	VEHICLE SAFETY SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To introduce the fundamentals and principles of vehicle safety and safety engineering.	
2.	To provide knowledge of braking systems and their components used in modern vehicles.	
3.	To explain stabilization systems for improving vehicle handling and safety.	
4.	To analyze braking and stabilization systems used in commercial vehicles.	
5.	To develop understanding of occupant safety, airbag systems, and driver distraction prevention strategies.	
UNIT–I	FUNDAMENTALS OF VEHICLE SAFETY	9
	Concepts of vehicle safety and design for uncertainty – Cause effect analysis - safety factors - component safety factor-Digital models and human testing - Safety compliance and regulatory considerations.	
UNIT–II	BRAKING SYSTEMS: DESIGN AND COMPONENTS	9
	Principles of braking systems - Brake system components - brake circuit configurations - Braking system design requirements and performance - Braking systems for passenger cars and light utility vehicles - Brake booster - master cylinder - Braking force limiters - Disc and drum brakes.	
UNIT– III	VEHICLE STABILIZATION SYSTEMS (PASSENGER CARS)	9
	Anti-lock braking system (ABS)- Traction control system (TCS) - Electronic stability program (ESP) - Electro-hydraulic brakes - Performance and safety implications.	
UNIT – IV	BRAKING AND STABILIZATION IN COMMERCIAL VEHICLES	9
	raking system configurations - Air supply and processing systems - Transmission devices and wheel brakes - Parking brake and retarder systems - Stabilization systems: ESP for commercial vehicles - Electronically controlled braking (ELB) - Electro-pneumatic braking: function - components - system design.	
UNIT–V	OCCUPANT INJURY PREVENTION AND DRIVER SAFETY	9
	Principles of occupant injury prevention - Proper use of head restraints - Airbags and their deployment systems - Distracted driving: risk assessment - information processing - Strategies for risk reduction.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Explain safety principles, design uncertainty, and compliance requirements in vehicles.	L2 - Understand
CO2	Describe braking system components and analyze braking performance	L2 -

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	in passenger vehicles.	Understand
CO3	Evaluate vehicle stabilization technologies such as ABS, TCS, and ESP.	L3 - Apply
CO4	Analyze braking and stabilization systems used in commercial vehicles.	L4 - Analyze
CO5	Assess occupant protection systems and strategies for reducing driver distraction.	L3 - Apply
	REFERENCE BOOKS:	
1.	George A. Peters, Barbara J. Peters, "Automotive vehicle safety", Taylor and Francis, 3rd Edition, 2015.	
2.	Robert Bosch GmbH, "Automotive Handbook", Wiley, 9th Edition, 2015.	
3.	Bimal K Bose, "Power Electronics and Motor Drive: Advances and Trends", Elsevier, 2006.	
4.	Thomas D. Gillespie, "Fundamentals of Vehicle Dynamics", SAE International, 1992.	
5.	T. Denton, "Automobile Mechanical and Electrical Systems", Routledge, 2017.	

Mapping of COs with POs

COs	PO1	PO2	PO3	PO4	PO5	PO6
CO1	2	1	3	2	2	1
CO2	2	1	3	3	1	1
CO3	2	1	3	3	2	2
CO4	3	2	3	3	2	2
CO5	2	1	2	2	3	1
Avg.	**2.2**	**1.2**	**2.8**	**2.6**	**2.0**	**1.4**
1-Low- 2 -Medium- 3-High.						

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ME23AE416	HUMAN FACTORS AND ERGONOMICS IN AUTOMOTIVE DESIGN	CP	L	T	P	C
		3	3	0	0	3
Programme & Branch	M.E. AUTOMOTIVE ELECTRONICS	Version: 1.0				

Course Objectives:		
1.	To understand human physical, cognitive, and sensory characteristics relevant to vehicle design.	
2.	To study ergonomic principles applied to automotive interiors, controls, and displays.	
3.	To analyze driver behavior, workload, comfort, and safety considerations.	
4.	To apply human-machine interaction principles in modern automotive systems.	
5.	To incorporate human factors into safe, comfortable, and user-centered vehicle design.	
UNIT-I	INTRODUCTION TO HUMAN FACTORS AND ERGONOMICS	9
	Human factors and ergonomics – Scope and objectives – Importance in automotive design – Human-centered design philosophy – Anthropometry and biomechanics – Static and dynamic anthropometric data – Reach envelopes – Posture analysis – Application of anthropometric data in vehicle packaging.	
UNIT-II	DRIVER PHYSIOLOGY, PERCEPTION AND COGNITION	9
	Human sensory systems – Vision - hearing- touch - proprioception – Visual acuity - contrast sensitivity - night vision – Auditory perception and noise effects – Information processing model – Attention – perception – memory – decision making – Mental workload – Driver stress and fatigue.	
UNIT- III	AUTOMOTIVE ERGONOMICS AND VEHICLE INTERIOR DESIGN	9
	Seating design – Seat comfort – adjustability - posture and lumbar support – Steering wheel – pedals - control layout – Cabin layout - packaging – Visibility requirements – Mirror design and field of view – Climate comfort – Noise vibration harshness (NVH) considerations.	
UNIT – IV	HUMAN–MACHINE INTERACTION AND DRIVER ASSISTANCE SYSTEMS	9
	Human–machine interface (HMI) concepts – Instrument clusters and displays – Touchscreens - switches - voice interfaces – Usability principles – Multimodal interaction – Driver distraction and workload management – Ergonomic evaluation of ADAS – Trust and acceptance of automation.	
UNIT–V	SAFETY, COMFORT AND ERGONOMIC EVALUATION METHODS	9
	Human factors in vehicle safety – Driver error and accident causation – Passive and active safety systems – Crashworthiness and occupant protection – Ergonomic assessment tools – Digital human modeling – Standards and regulations – User trials and ergonomic validation.	
	Total (L)	45 Periods
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: **Upon completion of this course the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply human factors and ergonomic principles in automotive vehicle design.	L3 - Apply
CO2	Analyze driver physiological and cognitive characteristics affecting driving performance.	L4 - Analyze
CO3	Design ergonomic vehicle interiors considering comfort, visibility, and control layout.	L3 - Apply
CO4	Apply human-machine interaction principles in automotive displays and driver assistance systems.	L3 - Apply
CO5	Evaluate vehicle safety and comfort using ergonomic assessment methods and standards.	L3 - Apply
	REFERENCE BOOKS:	
1.	Wickens, C. D., Gordon, S. E., & Liu, Y., "An Introduction to Human Factors Engineering", Pearson Education, 2nd Edition, 2004.	
2.	Sanders, M. S., McCormick, E. J., "Human Factors in Engineering and Design", McGraw-Hill Education, 7th Edition, 1993.	
3.	Green, P., "Automotive Ergonomics: Driver-Vehicle Interaction", SAE International, 1st Edition, 1999.	
4.	Porter, J. M., Case, K., Bonney, M. C., "Human Factors in Industrial and Product Design", Taylor & Francis, 1st Edition, 2004.	
5.	"SAE Human Factors Engineering Standards and Recommended Practices", SAE International, Latest Edition, 2020.	

Mapping of COs with POs

COs	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**
CO1	2	1	3	2	2	1
CO2	2	1	3	3	1	1
CO3	2	1	3	3	2	2
CO4	3	2	3	3	2	2
CO5	2	1	2	2	3	1
Avg.	**2.2**	**1.2**	**2.8**	**2.6**	**2.0**	**1.4**
1-Low- 2 -Medium- 3-High.						

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ME23IS501	ENVIRONMENTAL SAFETY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)	Version: 1.0				

Course Objectives:

1.	To provide in depth knowledge in Principles of Environmental safety and its applications in various fields.
2.	To give understanding of air and water pollution and their control.
3.	To expose the students to the basis in hazardous waste management.
4.	To provide knowledge on pollution monitoring and control devices.
5.	To design emission measurement devices.

UNIT–I	AIR POLLUTION	9
	Classification and properties of air pollutants -Pollution sources -Effects of air pollutants on human beings, Animals, Plants, and Materials -Automobile pollution -Hazards of air pollution -Concept of clean coal combustion technology -Ultra violet radiation, infrared radiation, radiation from the sun-Hazards due to depletion of ozone-Deforestation, ozone holes, automobile exhausts, chemical factory stack emissions, CFC.	
UNIT–II	WATER POLLUTION	9
	Classification of water pollutants-Health hazards-Sampling and analysis of water-Water treatment -Different industrial effluents and their treatment and disposal-Advanced wastewater treatment-Effluent quality standards and laws-Chemical industries, tannery, textile effluents -Common treatment.	
UNIT–III	HAZARDOUS WASTE MANAGEMENT	9
	Hazardous waste management in India-Waste identification, characterization, and classification-Technological options for collection, treatment and disposal of hazardous waste, Selection charts for the treatment of different hazardous wastes-Methods of collection and disposal of solid wastes-Health hazards -Toxic and radioactive wastes-Incineration and vitrification-Hazards due to bio-process, dilution, standards, and restrictions-Recycling and reuse.	
UNIT–IV	ENVIRONMENTAL MEASUREMENT AND CONTROL	9
	Sampling and analysis-Dust monitor -Gas analyzer- particle size analyzer-Lux meter- pH meter-Gas chromatograph-Atomic absorption spectrometer-Gravitational settling chambers, cyclone separators, scrubbers-Electrostatic precipitator, bag filter, maintenance-Control of gaseous emission by adsorption, absorption, and combustion methods-Pollution Control Board, laws.	

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UNIT–V	POLLUTION CONTROL IN PROCESS INDUSTRIES	9
	Pollution control in process industries -Cement, paper, petroleum, petroleum products, textile-Tanneries, thermal power plants-Dyeing and pigment industries -Eco-friendly energy.	
	Total(L)	45 Periods
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate and familiarize the basic concepts scope of environmental safety.	L2-Understand
CO2	Interpret the standards of professional conduct that are published by professional safety organizations and/or certification bodies.	L2- Understand
CO3	Explain the ways in which environmental health problems have arisen due to air and water pollution.	L2- Understand
CO4	Examine the role of hazardous waste management and use of critical thinking to identify and assess environmental health risks.	L4 - Analyze
CO5	Apply concepts of emission measurement and design emission measurement devices.	L3 - Apply

S.No.	REFERENCE BOOKS:
1.	E C Wolfe, "Race to Save to Save Planet", Wadsworth Publishing Co., Belmont, 2006.
2.	G T Miller, "Environmental Science: Working with the Earth", Wadsworth Publishing Co., Belmont, 11th Edition, 2006.
3.	M J Hammer and M J Hammer, "Water and Wastewater Technology", Pearson Prentice Hall, 2006
4.	Rao C S, "Environmental pollution Engineering", Wiley Eastern Limited, New Delhi, 2018.
5.	S P Mahajan, "Pollution control in process industries", Tata McGraw Hill Publishing Company, New Delhi, 2006.
6.	Varma and Braner, "Air pollution Equipment", Springer Publishers, Second Edition.

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ME23IS502	ELECTRICAL SAFETY	CP 3	L 3	T 0	P 0	C 3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)	Version: 1.0				

Course Objectives:

1.	To impart knowledge on fundamental electrical concepts, equipment principles, and comply with safety regulations, including basic first aid.
2.	To familiarize students with primary electrical hazards, insulation, and lightning protection measures.
3.	To provide an in depth knowledge on functioning of fuses, circuit breakers, and safety measures against electrical faults.
4.	To provide knowledge on equipment selection, safety features, and maintenance for electrical tools.
5.	To familiarize students with hazardous zone classification, safe equipment, and safety measures in different environments.

UNIT–I	**CONCEPTS AND STATUTORY REQUIREMENTS**	**9**
	Introduction – electrostatics, electro magnetism, stored energy, energy radiation and electromagnetic interference– Working principles of electrical equipment-Indian electricity act and rules-statutory requirements from electrical inspectorate-international standards on electrical safety– first aid-Cardio Pulmonary Resuscitation(CPR).	
UNIT–II	**ELECTRICAL HAZARDS**	**9**
	Primary and secondary hazards-shocks, burns, scalds, falls-human safety in the use of electricity. Energy leakage-clearances and insulation-classes of insulation-voltage classifications-excess energy current surges-Safety in handling of war equipment's-over current and short circuit current-heating effects of current-electromagnetic forces-corona effect-static electricity–definition, sources, hazardous conditions, control, electrical causes of fire and explosion-ionization, spark and arc ignition energy-national electrical safety code ANSI. Lightning, hazards, lightning arrestor, installation – earthing, specifications, earth resistance, earth pit maintenance.	
UNIT–III	**PROTECTION SYSTEMS**	**9**
	Fuse, circuit breakers and overload relays– protection against over voltage and under voltage– safe limits of amperage – voltage –safe distance from lines-capacity and protection of conductor-joints-and connections, overload and short circuit protection-no load protection-earth fault protection. FRLS insulation-insulation and continuity test-system grounding-equipment grounding-earth leakage circuit breaker (ELCB) -cable wires-maintenance of ground-ground fault circuit interrupter-use of low voltage-electrical guards-Personal protective equipment–safety in handling hand held electrical appliances tools and medical equipment's.	

UNIT–IV	**SELECTION, INSTALLATION, OPERATION AND MAINTENANCE**	**9**
	Role of environment in selection-safety aspects in application - protection and interlock-self diagnostic features and fail safe concepts-lock out and work permit system-discharge rod and earthing devices safety in the use of portable tools-cabling and cable joints-preventive maintenance.	
UNIT–V	**HAZARDOUS ZONES**	**9**
	Classification of hazardous zones-intrinsically safe and explosion proof electrical apparatus-increase safe equipment-their selection for different zones-temperature classification-grouping of gases-use of barriers and isolators-equipment certifying agencies.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Demonstrate understanding of electrical concepts and legal compliance for safe operation, within regulatory constraints.	L2 - Understand
CO2	Identify and mitigate electrical hazards, ensuring safety adherence to protocols and guidelines.	L3 - Apply
CO3	Utilize protection systems effectively, ensuring electrical safety within specified standards.	L3 - Apply
CO4	Apply a safe and efficient process for selecting, installing, operating, and maintaining electrical equipment, adhering to industry regulations.	L3 - Apply
CO5	Develop expertise in managing hazardous zones safely, within the constraints of applicable safety standards.	L3 - Apply
S.No.	**REFERENCE BOOKS:**	
1.	" Accident prevention manual for industrial operations", N.S.C., Chicago, 1982.	
2.	Indian Electricity Act and Rules, Government of India.	
3.	Power Engineers – Handbook of TNEB, Chennai, 1989.	
4.	Martin Glov, "Electrostatic Hazards in powder handling", Research Studies Pvt. Ltd., England, 1988.	
5.	Fordham Cooper W, "Electrical Safety Engineering", Butterworth and Company, London, 1986.	

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ME23IS503	SAFETY IN ENGINEERING INDUSTRY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To know the safety rules and regulations, standards and codes.
2.	To study various mechanical machines and their safety importance.
3.	To understand the principles of machine guarding and operation of protective devices.
4.	To know the working principle of mechanical engineering processes such as metal forming and joining process and their safety risks.
5.	To impart knowledge on finishing, inspection and testing operations in engineering industry.

UNIT–I	**SAFETY IN METAL WORKING MACHINERY AND WOOD WORKING MACHINES**	**9**
	General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines. Wood working machinery, types, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.	
UNIT–II	**PRINCIPLES OF MACHINE GUARDING**	**9**
	Guarding during maintenance, Zero Mechanical State (ZMS), Definition, Policy for ZMS– guarding of hazards- point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening. Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawing-shearing-presses-forge hammer -flywheels-shafts-couplings-gears-sprockets wheels and chains-pulleys and belts-authorized entry to hazardous installations-benefits of good guarding systems.	
UNIT–III	**SAFETY IN WELDING AND GAS CUTTING**	**9**
	Gas welding and oxygen cutting, resistances welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing–explosive welding, selection, care and maintenance of the associated equipment and instruments – safety in generation, distribution and handling of industrial gases-colour coding–flashback arrestor– leak detection-pipe line safety-storage and handling of gas cylinders.	

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UNIT–IV	SAFETY IN COLD FARMING AND HOT WORKING OF METALS	9
	Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance - metal sheers-press brakes. Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures. Safety in gas furnace operation, cupola, crucibles, ovens-foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes.	
UNIT–V	**SAFETY IN FINISHING, INSPECTION AND TESTING**	**9**
	Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation. Health and welfare measures in engineering industry-pollution control in engineering industry- industrial waste disposal.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply safety rules for maintaining and inspecting metal and wood working machines, ensuring industry standards.	L3 - Apply
CO2	Apply effective design strategies for machine guarding systems, emphasizing zero mechanical state (ZMS) during maintenance.	L3 - Apply
CO3	Demonstrate proficiency in safe welding and cutting, ensuring proper equipment selection, care, and maintenance.	L2 - Understand
CO4	Make use of safety measures in cold and hot metalworking, ensuring proper equipment setup, inspection, and maintenance.	L3 - Apply
CO5	Apply safety protocols in finishing, inspection, and testing, adhering to regulations and considering health and pollution control in engineering.	L3 - Apply
S.No.	**REFERENCE BOOKS:**	
1.	"Accident Prevention Manual", NSC, Chicago, 1982.	
2.	"Occupational safety Manual" BHEL, Trichy, 1988.	

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3.	John V. Grimaldi and Rollin H. Simonds, "Safety Management", All India Travelers Book seller, New Delhi, 1989.
4.	Krishnan N V, "Safety in Industry", Jaico Publishery House, 1996.
5.	Indian Boiler acts and Regulations, Government of India.
6.	"Safety in the use of wood working machines", HMSO, UK, 1992.
7.	"Health and Safety in welding and Allied processes", welding Institute, UK, High Tech. Publishing Ltd., London, 1989.

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Approved in Academic Council Meeting held on 13/12/2025

ME23IS504	DESIGN OF EXPERIMENTS	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				
Course Objectives:						
1.	To impart knowledge on principles and steps in designing a statistically designed experiment.					
2.	To build foundation in analysing the data in single factor experiments and to perform post hoc tests.					
3.	To provide knowledge on analysing the data in factorial experiments.					
4.	To educate on analysing the data analysis in special experimental designs and Response Surface Methods.					
5.	To impart knowledge in designing and analysing the data in Taguchi's Design of Experiments to improve Process/Product quality.					
UNIT–I	**EXPERIMENTAL DESIGN FUNDAMENTALS**	**9**				
	Importance of experiments, experimental strategies, basic principles of design, terminology, ANOVA, steps in experimentation, sample size, normal probability plot, linear regression models.					
UNIT–II	**SINGLE FACTOR EXPERIMENTS**	**9**				
	Completely randomized design, Randomized block design, Latin square design, Statistical analysis, estimation of model parameters, model adequacy checking, pair wise comparison tests.					
UNIT–III	**MULTIFACTOR EXPERIMENTS**	**9**				
	Two and three factor full factorial experiments, Randomized block factorial design, Experiments with random factors, rules for expected mean squares, approximate F-tests. 2^K factorial Experiments.					
UNIT–IV	**SPECIAL EXPERIMENTAL DESIGNS**	**9**				
	Blocking and confounding in 2^K designs. Two level Fractional factorial design, nested designs, Split plot design, Introduction to Response Surface Methods.					
UNIT–V	**TAGUCHI METHODS**	**9**				
	Steps in experimentation, design using Orthogonal Arrays, data analysis, Robust design,- control and noise factors, S/N ratios, parameter design, Multi-level experiments, Multi-response optimization, Introduction to Shainin DOE.					
	Total(L)	**45 Periods**				

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	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Interpret the Design of Experiments principles, strategizing experiment design within practical resource considerations and goals.	L2 - Understand
CO2	Analyze single-factor experiment data, focusing on randomization and pair-wise comparison tests.	L4 - Analyze
CO3	Analyze multifactor experiment data, applying rules for expected mean squares and approximate F-tests.	L4 - Analyze
CO4	Apply special experimental designs, minimize confounding effects, optimize data collection, and introduce Response Surface Methods with practical considerations.	L3 - Apply
CO5	Apply Taguchi-based approaches for quality evaluation, emphasizing practical experimentation with orthogonal arrays and multi-response optimization.	L3 - Apply

S.No.	**REFERENCE BOOKS:**
1.	Krishnaiah K and Shahabudeen P, "Applied Design of Experiments and Taguchi Methods", PHI learning private Ltd., 2012.
2.	Montgomery D C, "Design and Analysis of Experiments", John Wiley and Sons, Eighth edition, 2012.
3.	NicoloBelavendram, "Quality by Design: Taguchi techniques for industrial Experimentation", Prentice Hall, 1995.
4.	Phillip J Rose, "Taguchi techniques for quality Engineering", McGraw Hill, 1996.

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ME23IS505	CIRCULAR ECONOMY	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				

Course Objectives:	
1.	To equip graduates with circularity expertise for diverse national and international job opportunities.
2.	To develop skilled manpower and foster entrepreneurship in Circular Economy.
3.	To facilitate student-professional interactions for real-world exposure in technology, research, innovation, and circular business models.
4.	To inspire students to address circularity business needs and pursue Research and Development (R&D) and entrepreneurship.
5.	To cultivate environmentally conscious entrepreneurs through core competencies in environmental education and collaborative university-industry partnerships.

UNIT–I	**INTRODUCTION TO CIRCULAR ECONOMY**	**9**
	Linear Economy and its emergence, Economic and Ecological disadvantages of linear economy, Replacing Linear economy by Circular Economy, Development of Concept of Circular Economy, A differential - Linear Vs Circular Economy.	
UNIT–II	**CHARACTERISTICS OF CIRCULAR ECONOMY**	**9**
	Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loops.	
UNIT–III	**CIRCULAR DESIGN, INNOVATION AND ASSESSMENT**	**9**
	Zero waste: Waste Management in context of Circular Economy, Circular design, Research and innovation, LCA, Circular Business.	
UNIT–IV	**CASE STUDIES**	**9**
	Business models, Solid Waste Management / Wastewater, Plastics: A case study, EPR: polluters pay principle, Industrial symbiosis/ Eco-parks.	
UNIT–V	**LEGAL AND POLICY FRAMEWORK**	**9**
	Role of governments and networks, Sharing best practices, Universal circular economy policy goals, India and CE strategy, ESG.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Differentiate Circular Economy from Linear Economy and showcase its practical application.	L2 - Understand
CO2	Apply Circular Economy principles, incorporating material recovery and waste reduction to illustrate the Butterfly diagram and emphasize the loops within the circular system.	L3 - Apply
CO3	Apply circular design and innovation principles, assess sustainability in Circular Economy, and examine circular business models	L3 - Apply
CO4	Analyze case studies on circular economy from different fields and connect these cases to Circular Economy concepts professionally.	L4 - Analyze
CO5	Infer government roles, share best practices, and articulate Circular Economy policy goals, demonstrating expertise in legal frameworks with an ESG focus, especially in India.	L2 - Understand

S.No.	REFERENCE BOOKS:
1.	María-Laura Franco-García, Jorge Carlos Carpio-Aguilar, Hans Bressers, "Towards Zero Waste: Circular Economy Boost, Waste to Resources", Springer International Publishing, 2019.
2.	Marcello Tonelli and Nicolo Cristoni, "Strategic Management and the Circular Economy", Routledge, 2018.
3.	Sadhan Kumar Ghosh, "Circular Economy: Global Perspective", Springer, 2020.
4.	Stahel and Walter R, "The Circular Economy: A User's Guide", Routledge, 2019.
5.	Lerwen Liu, Seeram Ramakrishna, "An Introduction to Circular Economy", Springer Singapore, 2021.

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ME23IS501	ENVIRONMENTAL SAFETY	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				
Course Objectives:						
1.	To provide in depth knowledge in Principles of Environmental safety and its applications in various fields.					
2.	To give understanding of air and water pollution and their control.					
3.	To expose the students to the basis in hazardous waste management.					
4.	To provide knowledge on pollution monitoring and control devices.					
5.	To design emission measurement devices.					
UNIT–I	**AIR POLLUTION**	**9**				
	Classification and properties of air pollutants -Pollution sources - Effects of air pollutants on human beings, Animals, Plants, and Materials -Automobile pollution -Hazards of air pollution -Concept of clean coal combustion technology -Ultra violet radiation, infrared radiation, radiation from the sun-Hazards due to depletion of ozone-Deforestation, ozone holes, automobile exhausts, chemical factory stack emissions, CFC.					
UNIT–II	**WATER POLLUTION**	**9**				
	Classification of water pollutants-Health hazards-Sampling and analysis of water-Water treatment -Different industrial effluents and their treatment and disposal-Advanced wastewater treatment-Effluent quality standards and laws-Chemical industries, tannery, textile effluents -Common treatment.					
UNIT–III	**HAZARDOUS WASTE MANAGEMENT**	**9**				
	Hazardous waste management in India-Waste identification, characterization, and classification-Technological options for collection, treatment and disposal of hazardous waste, Selection charts for the treatment of different hazardous wastes-Methods of collection and disposal of solid wastes-Health hazards -Toxic and radioactive wastes-Incineration and vitrification-Hazards due to bio-process, dilution, standards, and restrictions-Recycling and reuse.					
UNIT–IV	**ENVIRONMENTAL MEASUREMENT AND CONTROL**	**9**				
	Sampling and analysis-Dust monitor -Gas analyzer- particle size analyzer-Lux meter- pH meter-Gas chromatograph-Atomic absorption spectrometer-Gravitational settling chambers, cyclone separators, scrubbers-Electrostatic precipitator, bag filter, maintenance-Control of gaseous emission by adsorption, absorption, and combustion methods-Pollution Control Board, laws.					

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UNIT–V	**POLLUTION CONTROL IN PROCESS INDUSTRIES**	**9**
	Pollution control in process industries -Cement, paper, petroleum, petroleum products, textile-Tanneries, thermal power plants-Dyeing and pigment industries -Eco-friendly energy.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate and familiarize the basic concepts scope of environmental safety.	L2-Understand
CO2	Interpret the standards of professional conduct that are published by professional safety organizations and/or certification bodies.	L2- Understand
CO3	Explain the ways in which environmental health problems have arisen due to air and water pollution.	L2- Understand
CO4	Examine the role of hazardous waste management and use of critical thinking to identify and assess environmental health risks.	L4 - Analyze
CO5	Apply concepts of emission measurement and design emission measurement devices.	L3 - Apply

S.No.	**REFERENCE BOOKS:**
1.	E C Wolfe, "Race to Save to Save Planet", Wadsworth Publishing Co., Belmont, 2006.
2.	G T Miller, "Environmental Science: Working with the Earth", Wadsworth Publishing Co., Belmont, 11th Edition, 2006.
3.	M J Hammer and M J Hammer, "Water and Wastewater Technology", Pearson Prentice Hall, 2006
4.	Rao C S, "Environmental pollution Engineering", Wiley Eastern Limited, New Delhi, 2018.
5.	S P Mahajan, "Pollution control in process industries", Tata McGraw Hill Publishing Company, New Delhi, 2006.
6.	Varma and Braner, "Air pollution Equipment", Springer Publishers, Second Edition.

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ME23IS502	ELECTRICAL SAFETY	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To impart knowledge on fundamental electrical concepts, equipment principles, and comply with safety regulations, including basic first aid.
2.	To familiarize students with primary electrical hazards, insulation, and lightning protection measures.
3.	To provide an in depth knowledge on functioning of fuses, circuit breakers, and safety measures against electrical faults.
4.	To provide knowledge on equipment selection, safety features, and maintenance for electrical tools.
5.	To familiarize students with hazardous zone classification, safe equipment, and safety measures in different environments.

UNIT–I	**CONCEPTS AND STATUTORY REQUIREMENTS**	**9**
	Introduction – electrostatics, electro magnetism, stored energy, energy radiation and electromagnetic interference– Working principles of electrical equipment-Indian electricity act and rules-statutory requirements from electrical inspectorate-international standards on electrical safety– first aid-Cardio Pulmonary Resuscitation(CPR).	
UNIT–II	**ELECTRICAL HAZARDS**	**9**
	Primary and secondary hazards-shocks, burns, scalds, falls-human safety in the use of electricity. Energy leakage-clearances and insulation-classes of insulation-voltage classifications-excess energy current surges-Safety in handling of war equipment's-over current and short circuit current-heating effects of current-electromagnetic forces-corona effect-static electricity–definition, sources, hazardous conditions, control, electrical causes of fire and explosion-ionization, spark and arc ignition energy-national electrical safety code ANSI. Lightning, hazards, lightning arrestor, installation – earthing, specifications, earth resistance, earth pit maintenance.	
UNIT–III	**PROTECTION SYSTEMS**	**9**
	Fuse, circuit breakers and overload relays– protection against over voltage and under voltage– safe limits of amperage – voltage –safe distance from lines-capacity and protection of conductor-joints-and connections, overload and short circuit protection-no load protection-earth fault protection. FRLS insulation-insulation and continuity test-system grounding-equipment grounding-earth leakage circuit breaker (ELCB) -cable wires-maintenance of ground-ground fault circuit interrupter-use of low voltage-electrical guards-Personal protective equipment–safety in handling hand held electrical appliances tools and medical equipments.	

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UNIT–IV	**SELECTION, INSTALLATION, OPERATION AND MAINTENANCE**	**9**
	Role of environment in selection-safety aspects in application - protection and interlock-self diagnostic features and fail safe concepts-lock out and work permit system-discharge rod and earthing devices safety in the use of portable tools-cabling and cable joints-preventive maintenance.	
UNIT–V	**HAZARDOUS ZONES**	**9**
	Classification of hazardous zones-intrinsically safe and explosion proof electrical apparatus-increase safe equipment-their selection for different zones-temperature classification-grouping of gases-use of barriers and isolators-equipment certifying agencies.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Demonstrate understanding of electrical concepts and legal compliance for safe operation, within regulatory constraints.	L2 - Understand
CO2	Identify and mitigate electrical hazards, ensuring safety adherence to protocols and guidelines.	L3 - Apply
CO3	Utilize protection systems effectively, ensuring electrical safety within specified standards.	L3 - Apply
CO4	Apply a safe and efficient process for selecting, installing, operating, and maintaining electrical equipment, adhering to industry regulations.	L3 - Apply
CO5	Develop expertise in managing hazardous zones safely, within the constraints of applicable safety standards.	L3 - Apply

S.No.	**REFERENCE BOOKS:**
1.	" Accident prevention manual for industrial operations", N.S.C., Chicago, 1982.
2.	Indian Electricity Act and Rules, Government of India.
3.	Power Engineers – Handbook of TNEB, Chennai, 1989.
4.	Martin Glov, "Electrostatic Hazards in powder handling", Research Studies Pvt. Ltd., England, 1988.
5.	Fordham Cooper W, "Electrical Safety Engineering", Butterworth and Company, London, 1986.

ME23IS503	SAFETY IN ENGINEERING INDUSTRY	CP 3	L 3	T 0	P 0	C 3
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To know the safety rules and regulations, standards and codes.
2.	To study various mechanical machines and their safety importance.
3.	To understand the principles of machine guarding and operation of protective devices.
4.	To know the working principle of mechanical engineering processes such as metal forming and joining process and their safety risks.
5.	To impart knowledge on finishing, inspection and testing operations in engineering industry.

UNIT–I	**SAFETY IN METAL WORKING MACHINERY AND WOOD WORKING MACHINES**	**9**
	General safety rules, principles, maintenance, Inspections of turning machines, boring machines, milling machine, planning machine and grinding machines, CNC machines. Wood working machinery, types, safety principles, electrical guards, work area, material handling, inspection, standards and codes- saws, types, hazards.	
UNIT–II	**PRINCIPLES OF MACHINE GUARDING**	**9**
	Guarding during maintenance, Zero Mechanical State (ZMS), Definition, Policy for ZMS– guarding of hazards- point of operation protective devices, machine guarding, types, fixed guard, interlock guard, automatic guard, trip guard, electron eye, positional control guard, fixed guard fencing- guard construction- guard opening. Selection and suitability: lathe-drilling-boring-milling-grinding-shaping-sawing-shearing-presses-forge hammer -flywheels-shafts-couplings-gears-sprockets wheels and chains-pulleys and belts-authorized entry to hazardous installations-benefits of good guarding systems.	
UNIT–III	**SAFETY IN WELDING AND GAS CUTTING**	**9**
	Gas welding and oxygen cutting, resistances welding, arc welding and cutting, common hazards, personal protective equipment, training, safety precautions in brazing, soldering and metalizing–explosive welding, selection, care and maintenance of the associated equipment and instruments – safety in generation, distribution and handling of industrial gases-colour coding–flashback arrestor– leak detection-pipe line safety-storage and handling of gas cylinders.	

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UNIT–IV	SAFETY IN COLD FARMING AND HOT WORKING OF METALS	9
	Cold working, power presses, point of operation safe guarding, auxiliary mechanisms, feeding and cutting mechanism, hand or foot-operated presses, power press electric controls, power press set up and die removal, inspection and maintenance - metal sheers-press brakes. Hot working safety in forging, hot rolling mill operation, safe guards in hot rolling mills – hot bending of pipes, hazards and control measures. Safety in gas furnace operation, cupola, crucibles, ovens-foundry health hazards, work environment, material handling in foundries, foundry production cleaning and finishing foundry processes.	
UNIT–V	SAFETY IN FINISHING, INSPECTION AND TESTING	9
	Heat treatment operations, electro plating, paint shops, sand and shot blasting, safety in inspection and testing, dynamic balancing, hydro testing, valves, boiler drums and headers, pressure vessels, air leak test, steam testing, safety in radiography, personal monitoring devices, radiation hazards, engineering and administrative controls, Indian Boilers Regulation. Health and welfare measures in engineering industry-pollution control in engineering industry- industrial waste disposal.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply safety rules for maintaining and inspecting metal and wood working machines, ensuring industry standards.	L3 - Apply
CO2	Apply effective design strategies for machine guarding systems, emphasizing zero mechanical state (ZMS) during maintenance.	L3 - Apply
CO3	Demonstrate proficiency in safe welding and cutting, ensuring proper equipment selection, care, and maintenance.	L2 - Understand
CO4	Make use of safety measures in cold and hot metalworking, ensuring proper equipment setup, inspection, and maintenance.	L3 - Apply
CO5	Apply safety protocols in finishing, inspection, and testing, adhering to regulations and considering health and pollution control in engineering.	L3 - Apply
S.No.	**REFERENCE BOOKS:**	
1.	"Accident Prevention Manual", NSC, Chicago, 1982.	
2.	"Occupational safety Manual", BHEL, Trichy, 1988.	

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3.	John V. Grimaldi and Rollin H. Simonds, "Safety Management", All India Travelers Book seller, New Delhi, 1989.
4.	Krishnan N V, "Safety in Industry", Jaico Publishery House, 1996.
5.	Indian Boiler acts and Regulations, Government of India.
6.	"Safety in the use of wood working machines", HMSO, UK, 1992.
7.	"Health and Safety in welding and Allied processes", welding Institute, UK, High Tech. Publishing Ltd., London, 1989.

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ME23IS504	DESIGN OF EXPERIMENTS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Mechanical Engineering					
Programme & Branch	Common to M.E. (AE, PED and SE)	Version: 1.0				

Course Objectives:

1.	To impart knowledge on principles and steps in designing a statistically designed experiment.
2.	To build foundation in analysing the data in single factor experiments and to perform post hoc tests.
3.	To provide knowledge on analysing the data in factorial experiments.
4.	To educate on analysing the data analysis in special experimental designs and Response Surface Methods.
5.	To impart knowledge in designing and analysing the data in Taguchi's Design of Experiments to improve Process/Product quality.

UNIT–I	**EXPERIMENTAL DESIGN FUNDAMENTALS**	9
	Importance of experiments, experimental strategies, basic principles of design, terminology, ANOVA, steps in experimentation, sample size, normal probability plot, linear regression models.	
UNIT–II	**SINGLE FACTOR EXPERIMENTS**	9
	Completely randomized design, Randomized block design, Latin square design, Statistical analysis, estimation of model parameters, model adequacy checking, pair wise comparison tests.	
UNIT–III	**MULTIFACTOR EXPERIMENTS**	9
	Two and three factor full factorial experiments, Randomized block factorial design, Experiments with random factors, rules for expected mean squares, approximate F-tests. 2^K factorial Experiments.	
UNIT–IV	**SPECIAL EXPERIMENTAL DESIGNS**	9
	Blocking and confounding in 2^K designs. Two level Fractional factorial design, nested designs, Split plot design, Introduction to Response Surface Methods.	
UNIT–V	**TAGUCHI METHODS**	9
	Steps in experimentation, design using Orthogonal Arrays, data analysis, Robust design,- control and noise factors, S/N ratios, parameter design, Multi-level experiments, Multi-response optimization, Introduction to Shainin DOE.	
	Total(L)	45 Periods

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	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Interpret the Design of Experiments principles, strategizing experiment design within practical resource considerations and goals.	L2 - Understand
CO2	Analyze single-factor experiment data, focusing on randomization and pair-wise comparison tests.	L4 - Analyze
CO3	Analyze multifactor experiment data, applying rules for expected mean squares and approximate F-tests.	L4 - Analyze
CO4	Apply special experimental designs, minimize confounding effects, optimize data collection, and introduce Response Surface Methods with practical considerations.	L3 - Apply
CO5	Apply Taguchi-based approaches for quality evaluation, emphasizing practical experimentation with orthogonal arrays and multi-response optimization.	L3 - Apply

S.No.	**REFERENCE BOOKS:**
1.	Krishnaiah K and Shahabudeen P, "Applied Design of Experiments and Taguchi Methods", PHI learning private Ltd., 2012.
2.	Montgomery D C, "Design and Analysis of Experiments", John Wiley and Sons, Eighth edition, 2012.
3.	NicoloBelavendram, "Quality by Design: Taguchi techniques for industrial Experimentation", Prentice Hall, 1995.
4.	Phillip J Rose, "Taguchi techniques for quality Engineering", McGraw Hill, 1996.

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ME23IS505	**CIRCULAR ECONOMY**	**CP**	**L**	**T**	**P**	**C**
		3	**3**	**0**	**0**	**3**
Type of Course	**Open Elective**					
Offering Dept.	**Mechanical Engineering**					
Programme & Branch	**Common to M.E. (AE, PED and SE)**	**Version: 1.0**				

Course Objectives:

1.	To equip graduates with circularity expertise for diverse national and international job opportunities.
2.	To develop skilled manpower and foster entrepreneurship in Circular Economy.
3.	To facilitate student-professional interactions for real-world exposure in technology, research, innovation, and circular business models.
4.	To inspire students to address circularity business needs and pursue Research and Development (R&D) and entrepreneurship.
5.	To cultivate environmentally conscious entrepreneurs through core competencies in environmental education and collaborative university-industry partnerships.

UNIT–I	**INTRODUCTION TO CIRCULAR ECONOMY**	**9**
	Linear Economy and its emergence, Economic and Ecological disadvantages of linear economy, Replacing Linear economy by Circular Economy, Development of Concept of Circular Economy, A differential - Linear Vs Circular Economy.	
UNIT–II	**CHARACTERISTICS OF CIRCULAR ECONOMY**	**9**
	Material recovery, Waste Reduction, reducing negative externalities, Explaining Butterfly diagram, Concept of Loops.	
UNIT–III	**CIRCULAR DESIGN, INNOVATION AND ASSESSMENT**	**9**
	Zero waste: Waste Management in context of Circular Economy, Circular design, Research and innovation, LCA, Circular Business.	
UNIT–IV	**CASE STUDIES**	**9**
	Business models, Solid Waste Management / Wastewater, Plastics: A case study, EPR: polluters pay principle, Industrial symbiosis/ Eco-parks.	
UNIT–V	**LEGAL AND POLICY FRAMEWORK**	**9**
	Role of governments and networks, Sharing best practices, Universal circular economy policy goals, India and CE strategy, ESG.	
	Total(L)	**45 Periods**
	OPEN-ENDEDPROBLEMS/QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End Semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Differentiate Circular Economy from Linear Economy and showcase its practical application.	L2 - Understand
CO2	Apply Circular Economy principles, incorporating material recovery and waste reduction to illustrate the Butterfly diagram and emphasize the loops within the circular system.	L3 - Apply
CO3	Apply circular design and innovation principles, assess sustainability in Circular Economy, and examine circular business models	L3 - Apply
CO4	Analyze case studies on circular economy from different fields and connect these cases to Circular Economy concepts professionally.	L4 - Analyze
CO5	Infer government roles, share best practices, and articulate Circular Economy policy goals, demonstrating expertise in legal frameworks with an ESG focus, especially in India.	L2 - Understand

S.No.	REFERENCE BOOKS:
1.	María-Laura Franco-García, Jorge Carlos Carpio-Aguilar, Hans Bressers, "Towards Zero Waste: Circular Economy Boost, Waste to Resources", Springer International Publishing, 2019.
2.	Marcello Tonelli and Nicolo Cristoni, "Strategic Management and the Circular Economy", Routledge, 2018.
3.	Sadhan Kumar Ghosh, "Circular Economy: Global Perspective", Springer, 2020.
4.	Stahel and Walter R, "The Circular Economy: A User's Guide", Routledge, 2019.
5.	Lerwen Liu, Seeram Ramakrishna, "An Introduction to Circular Economy", Springer Singapore, 2021.

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ME23PE501	ELECTRIC VEHICLE TECHNOLOGY	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)	Version: 1.0				

Course Objectives:

1.	To explain the historical evolution and the need for electric and hybrid vehicles.
2.	To describe electric vehicle architectures, powertrain layouts, and performance characteristics.
3.	To understand energy storage technologies, battery characteristics, and charging systems.
4.	To analyze different electric motors, power electronics drives, and control strategies used in EVs.
5.	To understand EV performance parameters, safety aspects, and major challenges.

UNIT–I	**NEED FOR ELECTRIC VEHICLES**	**9**
	History and Need for Electric and Hybrid Vehicles - Social and Environmental Importance of Hybrid and Electric Vehicles - Impact of Modern drive-trains on Energy Supplies - Comparison of Diesel, Petrol, Electric and Hybrid Vehicles – Limitations and Technical Challenges	
UNIT–II	**ELECTRIC VEHICLE ARCHITECHTURE**	**9**
	Electric Vehicle Types and Layout – Traction Motor Characteristics - Transmission Requirements - Concepts of Hybrid Electric Drive Train - Architecture of Series and Parallel Hybrid Electric Drive Train - Mild and Full Hybrids - Plug-in Hybrid Electric Vehicles.	
UNIT–III	**ENERGY STORAGE**	**9**
	Batteries – Types: Lead Acid Batteries, Nickel based Batteries, and lithium based batteries - Electrochemical Reactions - Energy Efficiency - Battery Charging Methods – Charger Types - Battery Cooling Methods.	
UNIT – IV	**ELECTRIC DRIVES AND CONTROL**	**9**
	Types of Electric Motors – Advantages and Limitations - DC Motor Drives and Control - Induction Motor Drives and Control - PMSM and Brushless DC motor - Speed Controllers.	
UNIT – V	**EV – PERFORMANCE AND CHALLENGES**	**9**
	Range, Efficiency, Acceleration, Gradeability, and Braking - Safety Aspects in Electric Vehicles – Challenges in Electric Vehicle Charging Infrastructure, Recycling, and Environmental Impact.	
	Total(L)	**45 Periods**
	*** INTERNAL EVALUATION ONLY**	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and	

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	evaluated as Internal Assessment only and not for the End semester Examinations.	
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand the need for electric vehicles, their benefits, and limitations compared to conventional vehicles.	L2 - Understand
CO2	Explain electric vehicle architecture, types, and hybrid drive-train concepts.	L2 - Understand
CO3	Understand different energy storage systems, battery operation, and charging methods.	L2 - Understand
CO4	Identify electric motors used in EVs and explain their control techniques.	L2 - Understand
CO5	Apply EV performance parameters techniques to solve safety issues, and key implementation challenges.	L3 - Apply
	REFERENCE BOOKS:	
1.	Iqbal Hussein, "Electric and Hybrid Vehicles: Design Fundamentals", 2nd Edition CRC Press, 2011.	
2.	Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, "Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design", CRC Press, 2004.	
3.	James Larminie, John Lowry, "Electric Vehicle Technology Explained", Wiley, 2003.	

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ME23PE502	ENERGY CONSERVATION AND MANAGEMENT	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)	Version: 1.0				

Course Objectives:

1.	To learn the present energy scenario and the need for energy conservation.
2.	To learn basics of refrigeration and air conditioning and methods to improve energy efficiency.
3.	To understand efficient lighting systems, sensors, and energy saving measures in appliances.
4.	To learn concepts of energy-efficient buildings, passive design, and renewable energy use.
5.	To understand the need, types, working, and applications of energy storage systems.

UNIT–I	**ENERGY SCENARIO**	**9**
	Primary Energy Resources - Sectorial Energy Consumption (Domestic, Industrial and other sectors) - Energy Pricing - Energy Conservation and its Importance - Energy Conservation Act-2001 and its features – Energy Star Rating.	
UNIT–II	**HVAC SYSTEMS**	**9**
	Basics of Refrigeration and Air Conditioning – COP / EER / SEC Evaluation – SPV System Design & Optimization for Solar Refrigeration.	
UNIT–III	**LIGHTING AND APPLIANCES**	**9**
	Specification of Luminaries – Types – Efficacy – Selection & Application – Time Sensors – Occupancy Sensors – Energy Conservation Measures in Electronic Devices.	
UNIT – IV	**ENERGY EFFICIENT BUILDINGS**	**9**
	Conventional versus Energy Efficient Buildings – Landscape Design – Envelope Heat Loss and Heat Gain – Passive Cooling and Heating – Renewable Sources Integration.	
UNIT – V	**ENERGY STORAGE TECHNOLOGIES**	**9**
	Necessity & Types of Energy Storage – Thermal Energy Storage – Battery Energy Storage, Charging and Discharging – Hydrogen Energy Storage & Super Capacitors – Energy Density and Safety Issues – Applications.	
	Total (L)	**45 Periods**
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Understand technical aspects of energy conservation scenario.	L2 - Understand
CO2	Understand the basics of refrigeration and HVAC systems and how they help in saving energy.	L2 - Understand
CO3	Learn how energy-efficient lighting and electronic appliances reduce power consumption.	L2 - Understand
CO4	Apply concepts of energy-efficient and sustainable building design.	L3 - Apply
CO5	Gain basic knowledge of different energy storage technologies and their applications.	L2 - Understand
	REFERENCE BOOKS:	
1.	Yogi Goswami, Frank Kreith, "Energy Efficiency and Renewable energy Handbook", CRC Press, 2016.	
2.	ASHRAE Handbook 2020 – "HVAC Systems & Equipment"	
3.	Paolo Bertoldi, Andrea Ricci, Anibal de Almeida, "Energy Efficiency in Household Appliances and Lighting", Springer, 2001.	
4.	David A. Bainbridge, Ken Haggard, Kenneth L. Haggard, "Passive Solar Architecture: Heating, Cooling, Ventilation, Daylighting, and More using Natural Flows", Chelsea Green Publishing, 2011.	
5.	Guide book for National Certification Examination for Energy Managers and Energy Auditors.	

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ME23PE503	SMART GRID	CP 3	L 3	T 0	P 0	C 3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)	Version: 1.0				

Course Objectives:

1.	To study about smart grid technologies, different smart meters and advanced metering infrastructure.	
2.	To know about the function of smart grid.	
3.	To familiarize the power quality management issues in smart grid.	
4.	To familiarize the high performance computing for smart grid applications	
5.	To get familiarized with the communication networks for smart grid applications	
UNIT–I	**INTRODUCTION**	9
	Evolution of Electric Grid - Concept, Definitions and Need for Smart Grid - Smart Grid Drivers, functions, Opportunities, Challenges and benefits - Difference between Conventional & Smart Grid - Smart Grid Initiative in India – Case Study.	
UNIT–II	**SMART GRID TECHNOLOGIES**	9
	Integration of Renewable Energy Sources - Smart substations, Substation Automation - Protection and control - Fault Detection, Isolation and service restoration - Outage management - Grid to Vehicle and Vehicle to Grid charging concepts.	
UNIT–III	**ADVANCED METERING INFRASTRUCTURE**	9
	Introduction to Smart Meters - Advanced Metering infrastructure (AMI) drivers and benefits - AMI protocols and standards - AMI needs in the Smart Grid - Demand side Management and Demand Response Programs, Demand Pricing and Time of Use, Real Time Pricing and Peak Time Pricing.	
UNIT – IV	**POWER QUALITY MANAGEMENT**	9
	Power Quality in Smart Grid - Power Quality Issues - Power Quality Conditioners for Smart Grid - Web Based Power Quality Monitoring - Power Quality Audit.	
UNIT – V	**HIGH PERFORMANCE COMPUTING**	*9
	Basics of Communication Systems used in Smart Grid - LAN, HAN, WAN, Broadband Over Power Line (BPL), IP based Protocols - Basics of Web Service and Cloud Computing - Cyber Security for Smart Grid.	
	Total (L)	**45 Periods**
	*** INTERNAL EVALUATION ONLY**	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain smart grid concepts, features, and Indian smart grid initiatives.	L2 - Understand
CO2	Describe smart grid technologies and renewable energy integration methods.	L2 - Understand
CO3	Understand AMI systems and demand response programs.	L2 - Understand
CO4	Apply mitigation methods to solve power quality issues in smart grids.	L3 - Apply
CO5	Understand communication systems, cloud computing, and cyber security used in smart grid applications.	L2 - Understand
	REFERENCE BOOKS:	
1.	Stuart Borlase "Smart Grid: Infrastructure, Technology and Solutions", CRC Press 2012.	
2.	James Momoh, "Smart Grid: Fundamentals of Design and Analysis", IEEE press, A John Wiley & Sons, Inc., Publication, 2012	
3.	Janaka Ekanayake, Nick Jenkins, Jianzhong Wu, Akihiko Yokoyama, "Smart Grid: Technology and Applications", Wiley, 2012.	
4.	Mini S. Thomas, John D McDonald, "Power System SCADA and Smart Grids", CRC Press, 2015	
5.	Kenneth C.Budka, Jayant G. Deshpande, Marina Thottan, "Communication Networks for Smart Grids", Springer, 2014	

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ME23PE504	SOLAR PV AND ENERGY STORAGE SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	M.E. – POWER ELECTRONICS AND DRIVES					
Programme & Branch	Common to M.E (ISE, AE and SE)	Version: 1.0				

Course Objectives:

1.	To understand sunlight characteristics, semiconductor basics, and the working of solar cells.
2.	To learn the design and components of stand-alone and hybrid solar PV systems.
3.	To understand grid-connected PV systems, safety, performance, economics, and smart grid integration.
4.	To understand the need for energy storage and different storage technologies for solar power.
5.	To learn various practical applications of solar energy in industry, transport, and communication.

UNIT–I	**INTRODUCTION**	9
	Characteristics of Sunlight – Semiconductors and P-N Junctions – Behavior of Solar Cells – Cell Properties – PV cell Interconnection	
UNIT–II	**STAND ALONE PV SYSTEM**	9
	Solar Modules – Storage Systems – Power Conditioning and Regulation – MPPT - Protection – Stand alone PV Systems Design – Hybrid PV Systems.	
UNIT–III	**GRID CONNECTED PV SYSTEMS**	9
	PV Systems in Buildings – Design Issues for Central Power Stations – safety – Economic aspect – Efficiency and Performance - Smart Grid Integration.	
UNIT – IV	**ENERGY STORAGE SYSTEMS**	9
	Impact of Intermittent Generation – Battery Energy Storage – Solar Thermal Energy Storage – Pumped Hydroelectric Energy Storage - Supercapacitors and Flywheel Energy Storage – Principles and Applications	
UNIT – V	**APPLICATIONS**	9
	Solar Desalination - Water Pumping – Battery Chargers – Solar Car – Direct-drive Applications – Space – Telecommunications.	
	Total(L)	**45 Periods**
	*** INTERNAL EVALUATION ONLY**	
	OPEN-ENDED PROBLEMS / QUESTIONS	
	Course specific Open Ended Problems will be solved during the classroom teaching. Such problems can be given as Assignments and evaluated as Internal Assessment only and not for the End semester Examinations.	

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	Course Outcomes: Upon completion of this course, the students will be able to:	BLOOM'S Taxonomy
CO1	Explain the working principles of solar cells and PV systems.	L2 - Understand
CO2	Design basic stand-alone and hybrid solar PV systems.	L3 - Apply
CO3	Understand grid-connected PV systems and their integration with smart grids.	L2 - Understand
CO4	Identify suitable energy storage systems for intermittent solar generation.	L2 - Understand
CO5	Apply solar energy concepts to real-world applications.	L3 - Apply
	REFERENCE BOOKS:	
1.	Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd., 2015.	
2.	Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa,1994.	
3.	Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011. 4	
4.	McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 5	

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ME23SE501	BLOCKCHAIN TECHNOLOGIES	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand the fundamental concepts and working principles of Blockchain technology.
2.	To explore various components, mechanisms, and architectures used in Blockchain systems.
3.	To examine real-world applications of Blockchain across different domains.
4.	To gain practical knowledge of private and public Blockchain networks.
5.	To learn the basics of smart contracts and understand how they are implemented in Blockchain platforms.

UNIT–I	INTRODUCTION OF CRYPTOGRAPHY AND BLOCKCHAIN	9
	Introduction to Blockchain - Blockchain Technology Mechanisms & Networks - Blockchain Origins - Objective of Blockchain - Blockchain Challenges - Transactions and Blocks - P2P Systems - Keys as Identity - Digital Signatures – Hashing - and public key cryptosystems - private vs. public Blockchain.	
UNIT–II	INTRODUCTION TO ETHEREUM	9
	Introduction to Bitcoin - The Bitcoin Network - The Bitcoin Mining Process - Mining Developments - Bitcoin Wallets - Decentralization and Hard Forks - Ethereum Virtual Machine (EVM) - Merkle Tree - Double-Spend Problem - Blockchain and Digital Currency - Transactional Blocks - Impact of Blockchain Technology on Cryptocurrency.	
UNIT–III	BUSINESS FORECASTING	9
	Introduction to Ethereum - Consensus Mechanisms – Metamask Setup - Ethereum Accounts – Transactions - Receiving Ethers - Smart Contracts.	
UNIT–IV	INTRODUCTION TO HYPERLEDGER AND SOLIDITY PROGRAMMING	9
	Introduction to Hyperledger - Distributed Ledger Technology & its Challenges - Hyperledger & Distributed Ledger Technology - Hyperledger Fabric - Hyperledger Composer. Solidity - Language of Smart Contracts - Installing Solidity & Ethereum Wallet - Basics of	

	Solidity - Layout of a Solidity Source File & Structure of Smart Contracts - General Value Types.	
UNIT–V	**BLOCKCHAIN APPLICATIONS**	9
	Internet of Things - Medical Record Management System - Domain Name Service and Future of Blockchain - Alt Coins.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand and explore the working of blockchain technology.	L2 – Understand
CO2	Implement the bitcoin mining process.	L3 – Apply
CO3	Develop and deploy simple smart contracts.	L3 – Apply
CO4	Apply the concept of solidity to build de-centralized apps on ethereum.	L3 – Apply
CO5	Develop applications on blockchain.	L3 – Apply
	REFERENCE BOOKS:	
1.	Imran Bashir, "Mastering Blockchain: Distributed Ledger Technology, Decentralization, and Smart Contracts Explained", Second Edition, Packt Publishing, 2018.	
2.	Narayanan, J. Bonneau, E. Felten, A. Miller, S. Goldfeder, "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press, 2016	
3.	Antonopoulos, "Mastering Bitcoin", O'Reilly Publishing, 2014.	
4.	Antonopoulos and G. Wood, "Mastering Ethereum: Building Smart Contracts and Dapps", O'Reilly Publishing, 2018.	
5.	D. Drescher, "Blockchain Basics", Apress, 2017.	

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ME23SE502	DEEP LEARNING	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand Deep Neural Networks for solving complex learning tasks.
2.	To implement CNN-based models such as R-CNN, Fast R-CNN, Faster R-CNN, and Mask R-CNN for object detection and recognition.
3.	To build and train Recurrent Neural Networks and apply NLP techniques using word embedding.
4.	To understand the internal structure, working mechanisms, and differences between LSTM and GRU units.
5.	To apply Autoencoders for image processing, feature extraction, and representation learning.

UNIT–I	DEEP LEARNING CONCEPTS	9
	Fundamentals about Deep Learning - Perception Learning Algorithms - Probabilistic modelling - Early Neural Networks - How Deep Learning different from Machine Learning – Scalars - Vectors – Matrixes - Higher Dimensional Tensors - Manipulating Tensors- Vector Data- Time Series Data - Image Data - Video Data.	
UNIT–II	NEURAL NETWORKS	9
	About Neural Network - Building Blocks of Neural Network. Optimizers - Activation Functions - Loss Functions - Data Pre - processing for neural networks - Feature Engineering - Overfitting and Underfitting - Hyperparameters.	
UNIT–III	CONVOLUTIONAL NEURAL NETWORK	9
	CNN Basics: Linear Time Invariant, Image Processing & Filtering. Network Architecture: Input, Convolution, Pooling, Dense, Dropout, Batch Normalization Layers; Activation Functions; Optimizers; Backpropagation. Pretrained Models & Transfer Learning: LeNet, AlexNet, VGG16, ResNet, Inception, Google Inception, Microsoft ResNet. Object Detection: R-CNN, Fast R-CNN, Faster R-CNN, Mask R-CNN, YOLO.	
UNIT–IV	NATURAL LANGUAGE PROCESSING USING RNN	9
	About NLP & its Toolkits - Language Modeling - Vector Space Model (VSM) - Continuous Bag of Words (CBOW) - Skip-Gram Model for Word Embedding - Part of Speech (PoS) Global Co-	

	occurrence Statistics – based Word Vectors - Transfer Learning - Word2Vec - Global Vectors for Word Representation GloVe - Backpropagation Through Time - Bidirectional RNNs (BRNN) - Long Short Term Memory (LSTM) - Bi-directional LSTM - Sequence-to-Sequence Models (Seq2Seq) - Gated recurrent unit GRU.	
UNIT–V	**DEEP REINFORCEMENT & UNSUPERVISED LEARNING**	**9**
	About Deep Reinforcement Learning. Q-Learning - Deep Q-Network (DQN). Policy Gradient Methods - Actor-Critic Algorithm - About Autoencoding - Convolutional Auto Encoding - Variational Auto Encoding. Generative Adversarial Networks - Autoencoders for Feature Extraction - Auto Encoders for Classification - Denoising Autoencoders - Sparse Autoencoders.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Apply feature extraction techniques to derive meaningful representations from image and video datasets.	L3 – Apply
CO2	Apply neural network preprocessing and tuning techniques to build effective learning models.	L3 – Apply
CO3	Implement image recognition and image classification using a pretrained network.	L3 – Apply
CO4	Apply text mining and analytics techniques on Twitter data.	L3 – Apply
CO5	Implement autoencoders for classification and feature extraction.	L3 – Apply

	REFERENCE BOOKS:
1.	Josh Patterson and Adam Gibson, "Deep Learning A Practitioner's Approach", O'Reilly Media, Inc.2017.
2.	Jojo Moolayil ,"Learn Keras for Deep Neural Networks", Apress,2018.
3.	Vinita Silaparasetty, "Deep Learning Projects Using TensorFlow 2", Apress, 2020.
4.	Francois Chollet, "Deep Learning with Python", Manning Shelter Island, 2017.
5.	Santanu Pattanayak, "Pro Deep Learning with TensorFlow", Apress, 2017.

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ME23SE503	FULL STACK WEB APPLICATION DEVELOPMENT	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To develop typescript application.
2.	To develop single page application.
3.	To enable to communicate with a server over the HTTP protocol.
4.	To learn all the tools need to start building applications with Node.js.
5.	To implement the full stack development using MEAN Stack.

UNIT–I	FUNDAMENTALS & TYPESCRIPT LANGUAGE	9
	Server-Side Web Applications - Client-Side Web Applications - Single Page Application - About TypeScript - Creating TypeScript Projects - TypeScript Data Types – Variables - Expression and Operators – Functions - OOP in Typescript – Interfaces – Generics – Modules – Enums – Decorators – Enums – Iterators - Generators.	
UNIT–II	ANGULAR	9
	Angular Overview: Angular CLI - Project Creation - Components & Interaction - Dynamic Components - Angular Elements. Forms: Template - driven & Reactive Forms - Property/Style/Class/Event Binding - Two-way Binding – FormGroup – FormControl - Routing: Router Configuration – Navigation – RouterLink - Query Parameters - URL Matching Strategies. Services & HTTP: Dependency Injection – HttpClient - CRUD Operations - Http Headers - Request/Response.	
UNIT–III	NODE.Js	9
	Node.js Overview: Environment Setup – NPM – Modules - Asynchronous Programming - Callbacks & Error Handling - Call Stack & Event Loop. File System & I/O: Synchronous vs Asynchronous I/O - File Handling - Path/Directory Operations - Timers & Timer Promises API. Events & Buffers: EventEmitter - Event API - Buffers & TypedArrays - Binary Data Handling - Flowing vs Non-Flowing Streams. Data Handling: JSON Processing.	

UNIT–IV	**EXPRESS.Js**	**9**
	Introduction to Express.js - Configuring Express.js App Settings - Defining Routes - Starting the App - Express.js Application Structure – Configuration - Settings – Middleware - body-parser - cookie-parser - express-session - response-time - Template Engine – Jade – EJS – Parameters – Routing - router.route(path) - Router Class - Request Object - Response Object - Error Handling - RESTful.	
UNIT–V	**MONGODB**	**9**
	Introduction to MongoDB – Documents – Collections- Subcollections – Database - Data Types – Dates- Arrays - Embedded Documents - CRUD Operations - Batch Insert - Insert Validation - Querying The Documents – Cursors – Indexing - Unique Indexes - Sparse Indexes - Special Index and Collection Types - Full-Text Indexes- Geospatial Indexing - Aggregation framework.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Develop basic programming skills using javascript.	L3 – Apply
CO2	Implement a front-end web application using Angular.	L3 – Apply
CO3	Apply Node.js environment setup and NPM to create and manage modular javascript applications.	L3 – Apply
CO4	Apply Express.js RESTful principles to develop server-side web applications.	L3 – Apply
CO5	Implement the aggregation framework to process and analyze complex datasets in MongoDB.	L3 – Apply

	REFERENCE BOOKS:
1.	Adam Freeman, "Essential TypeScript", Apress, 2019.
2.	Mark Clow, "Angular Projects", Apress, 2018.
3.	Alex R. Young, Marc Harter,"Node.js in Practice", Manning Publication, 2014.
4.	Azat Mardan, "Pro Express.js", Apress, 2015.
5.	Kyle Banker, Peter Bakkum, Shaun Verch, Douglas Garrett, Tim Hawkins, "MongoDB in Action", Manning Publication, Second edition, 2016.

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ME23SE504	PRINCIPLES OF MULTIMEDIA	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand the gamut of multimedia and its significance.
2.	To understand the components and characteristics of multimedia systems.
3.	To gain familiarity with multimedia tools and authoring environments.
4.	To develop skills in designing and implementing multimedia applications.
5.	To explore the latest trends and technologies in multimedia.

UNIT–I	**INTRODUCTION**	**9**
	Introduction to Multimedia – Characteristics of Multimedia Presentation – Multimedia Components – Promotion of Multimedia Based Components – Digital Representation – Media and Data Streams – Multimedia Architecture – Multimedia Documents, Multimedia Tasks and Concerns, Production, sharing and distribution, Hypermedia, WWW and Internet, Authoring, Multimedia over wireless and mobile networks.	
UNIT–II	**ELEMENTS OF MULTIMEDIA**	**9**
	Text-Types, Font, Unicode Standard, File Formats, Graphics and Image data representations – data types, file formats, color models; video – color models in video, analog video, digital video, file formats, video display interfaces, 3D video and TV: Audio – Digitization, SNR, SQNR, quantization, audio quality, file formats, MIDI; Animation- Key Frames and Tweening - other Techniques, 2D and 3D Animation.	
UNIT–III	**MULTIMEDIA TOOLS**	**9**
	Authoring Tools – Features and Types – Card and Page Based Tools – Icon and Object Based Tools – Time Based Tools – Cross Platform Authoring Tools – Editing Tools – Painting and Drawing Tools – 3D Modeling and Animation Tools – Image Editing Tools – Sound Editing Tools – Digital Movie Tools.	
UNIT–IV	**MULTIMEDIA SYSTEMS**	**9**
	Compression Types and Techniques: CODEC, Text Compression: GIF Coding Standards, JPEG standard – JPEG 2000, basic audio compression – ADPCM, MPEG Psychoacoustics, basic Video	

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	compression techniques – MPEG, H.26X – Multimedia Database System – User Interfaces – OS Multimedia Support – Hardware Support – Real Time Protocols – Play Back Architectures – Synchronization – Document Architecture – Hypermedia Concepts: Hypermedia Design – Digital Copyrights, Content analysis.	
UNIT–V	**MULTIMEDIA APPLICATIONS FOR THE WEB AND MOBILE PLATFORMS**	**9**
	ADDIE Model – Conceptualization – Content Collection – Storyboard-Script Authoring Metaphors – Testing – Report Writing – Documentation - Multimedia for the web and mobile platforms. Virtual Reality - Internet multimedia content distribution - Multimedia Information sharing – social media sharing - cloud computing for multimedia services, interactive cloud gaming - Multimedia information retrieval.	
	Total (L)	**45 Periods**
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Illustrate multimedia architectures and their role in hypermedia and mobile networks.	L2 – Understand
CO2	Explain the key concepts and techniques used in multimedia applications.	L2 – Understand
CO3	Develop effective strategies to deliver quality of experience in multimedia applications.	L3 – Apply
CO4	Design and implement algorithms and techniques applied to multimedia objects.	L3 – Apply
CO5	Apply the ADDIE model and multimedia technologies for web, mobile, and cloud-based platforms.	L3 – Apply
	REFERENCE BOOKS:	
1.	Li, Ze-Nian, Drew, Mark, Liu, Jiangchuan, "Fundamentals of Multimedia", Third Edition, Springer, 2021.	
2.	Prabhat K.Andleigh, Kiran Thakrar, "Multimedia Systems Design", Pearson Education, 2015.	
3.	Gerald Friedland, Ramesh Jain, "Multimedia Computing", Cambridge University Press, 2018. (digital book)	
4.	Ranjan Parekh, "Principles of Multimedia", Second Edition, McGraw-Hill Education, 2017	

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ME23SE505	CYBER PHYSICAL SYSTEMS	CP	L	T	P	C
		3	3	0	0	3
Type of Course	Open Elective					
Offering Dept.	Computer Science and Engineering					
Programme & Branch	Common to M.E. (AE,PED,ISE)	Version: 1.0				

Course Objectives:

1.	To understand the design of cyber-physical systems advanced techniques for modeling and analysis.
2.	To explore cyber-physical systems from attacks by understanding and improving their security.
3.	To design and analyze synchronization techniques for reliable coordination in distributed cyber-physical systems.
4.	To apply real-time scheduling methods that handle fixed timing, memory effects, multicore systems, and system variability in cyber-physical systems.
5.	To learn techniques for modeling, integrating, and formalizing semantics of cyber-physical systems using domain-specific languages.

UNIT–I	**SYMBOLIC SYNTHESIS FOR CYBER-PHYSICAL SYSTEMS**	**9**
	Introduction and Motivation, Basic Techniques - Preliminaries, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, Advanced Techniques: Construction of Symbolic Models, Continuous-Time Controllers, Software Tools.	
UNIT–II	**SECURITY OF CYBER-PHYSICAL SYSTEMS**	**9**
	Introduction and Motivation - Basic Techniques - Cyber Security Requirements - Attack Model – Countermeasures - Advanced Techniques: System Theoretic Approaches.	
UNIT–III	**SYNCHRONIZATION IN DISTRIBUTED CYBER-PHYSICAL SYSTEMS**	9
	Challenges in Cyber-Physical Systems - A Complexity-Reducing Technique for Synchronization - Formal Software Engineering - Distributed Consensus Algorithms - Synchronous Lockstep Executions - Time-Triggered Architecture - Related Technology - Advanced Techniques.	
UNIT–IV	**REAL-TIME SCHEDULING FOR CYBER-PHYSICAL SYSTEMS**	**9**
	Introduction and Motivation - Basic Techniques - Scheduling with Fixed Timing Parameters - Memory Effects - Multiprocessor/Multicore Scheduling - Accommodating Variability and Uncertainty.	

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UNIT–V	MODEL INTEGRATION IN CYBER-PHYSICAL SYSTEMS	9
	Introduction and Motivation - Causality - Semantic Domains for Time - Interaction Models for Computational Processes - Semantics of CPS DSMLs - Advanced Techniques – ForSpec - The Syntax of CyPhyML - Formalization of Semantics - Formalization of Language Integration.	
	Total (L)	45 Periods
	Course Outcomes: **Upon completion of this course, the students will be able to:**	**BLOOM'S Taxonomy**
CO1	Understand the core principles behind Cyber physical systems.	L2 – Understand
CO2	Identify Security mechanisms of Cyber physical systems.	L2 – Understand
CO3	Understand Synchronization in Distributed Cyber-Physical Systems.	L2 – Understand
CO4	Apply basic scheduling techniques with fixed timing parameters.	L3 -Apply
CO5	Apply the semantics of Cyber physical Domain-Specific Modeling Languages to model system behavior accurately.	L3 -Apply
	REFERENCE BOOKS:	
1.	Raj Rajkumar, Dionisio De Niz, and Mark Klein, "Cyber-Physical Systems", 1st edition, Addison-Wesley Professional, 2017.	
2.	Rajeev Alur, "Principles of Cyber-Physical Systems", MIT Press, 2015.	

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